



Project Task 6a (Group Project)

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Abstract

This project focuses on designing and implementing a robust network infrastructure to support the Faculty of Computing's anticipated growth and technological advancements. With an expected increase of 15% in students and staff over the next four years, the faculty requires an upgraded two-story building featuring multiple computing labs, a hybrid classroom, a video conferencing room and a student lounge. The objective of this project is to develop a high-performance, scalable and secure network that facilitates seamless connectivity, efficient data management and enhanced digital learning experiences.

The project involves several key tasks, including network planning, hardware selection, LAN and WAN configuration, IP addressing and security implementation. A hybrid network topology combining star and mesh structures is chosen to balance reliability, performance and cost-effectiveness. Enterprise-grade devices such as routers, switches, access points and fiber-optic cabling are integrated to ensure high-speed wired and wireless connections. Security measures, including VLAN segmentation, Quality of Service (QoS) and disaster recovery planning are incorporated to protect the network from potential threats and ensure uninterrupted operations.

Throughout the project, careful feasibility assessments were conducted to evaluate technical, economic and operational considerations. Task delegation, collaborative decision-making and iterative improvements played crucial roles in refining the final network design. The implementation of modern technologies such as Wi-Fi 6, VPN encryption and advanced monitoring tools ensures the infrastructure is future-proof and aligned with the evolving needs of digital education. This project highlights the importance of structured network design and strategic resource allocation in developing an optimized, secure and efficient computing environment.

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2. Introduction

As the Faculty of Computing (FC) might anticipate a 15% growth in students and academic staff over the next four years, there is a need to establish an upgraded infrastructure that can accommodate this growth. To support this expansion, FC plans to construct a new two-storey building that will house multiple laboratories, a hybrid classroom, Video Conferencing Room, and a student lounge, all requiring a robust and scalable network infrastructure. The aim of this project is to develop a high-performance network that ensures seamless connectivity, strong security, and ease of management while maintaining cost-effectiveness. The network must also incorporate high-speed wired and wireless access, scalable network architecture, optimized network routes to support modern educational needs in alignment with the Fourth Industrial Revolution (4IR).

The scope of this project includes planning and designing the network infrastructure, structuring network architecture, integrating advanced technology into the labs and rooms, budgeting, implementing the network and calculating network routes to ensure optimal performance. The objectives of our team are to establish efficient communication across all facilities, enhance network security, and ensure scalability for future expansion. By leveraging both our knowledge and advancements in networking technologies, we aim to create an innovative, connected educational environment that not only fulfils the current demands of FC but also positions it for long-term technological advancements and growth.

3. Project Background and an overview of the client's status and issue

The Faculty of Computing (FC) is currently facing a significant challenge as FC is projected to have 15% increase in students and staff population over the next four years. To accommodate this expansion, FC plans to construct a new two-storey building that will house 2 general purpose labs, a Cisco Network lab and an Embedded lab, a video conferencing room, a student lounge and hybrid classroom. Each lab and the student lounge will be 14mx10m in size and all interconnected via Wi-Fi. However, the successful implementation of this new facility depends on a well-designed, high-performance network infrastructure that can support the faculty's evolving technological and educational needs. As education moves towards digital learning and IoT-integrated environments, FC requires a scalable, secure, and efficient network system that enhances connectivity while being cost-effective.

Currently, FC need a better equipment that will allow for this growth. All stakeholders want to prepare for future needs to the extent possible at the best value possible which particularly ensure a reliable, efficient and security vulnerable network infrastructure that hinder its ability to support modern digital learning tools. We have to make sure the network is easy to manage and scalable, capable of enhancing overall performance, and equipped with robust security measures to protect against cyber threats, including internet worms, denial-of-service attacks, and e-business application vulnerabilities. Additionally, the network must provide high-performance support to the core backbone, ensuring uninterrupted access to critical resources such as virtual learning platforms, hybrid applications, and workshops. The transition from the current infrastructure to the upgraded system must be seamless, ensuring that users continue to have uninterrupted access throughout the implementation phase.

In short, the development of this new 2-storey building, featuring different labs with a strong focus on network infrastructure is designed to support the evolving needs of FC's academic community while maintaining a balance between innovation, long-term scalability, and cost efficiency for the future.

4. Compiled Solution

4.1 Task 1

4.1.1 Brief Explanation for The Task

Task 1 which is project setup focuses on creating a team and establishing the foundation setup for a small network design project. This project is to accommodate the growth of the students over the next 20 years, as the computing field grows and moves forward into the future. We formed our group of four students, including Chau Ying Jia, Poh Lok Yee, Sabrina Heng Wei Qi and Woo Cheng Shuan. We selected 4Fabs as our group name. The team's objective is to understand the case study and design a new 2-storey building. This building must house 4 labs which is 2 general purpose labs, 1 Cisco Network lab and 1 Embedded lab. Besides, it must also include 1 video conferencing room, 1 hybrid classroom and a student lounge. Additionally, our group members will document all group meetings in detailed minutes, outlining discussions, decisions and responsibilities.

Group members have to research and based on each room different purpose, design in detail including what equipment and the fair positioning of them in the rooms. The floorplan is also part of our task to design a building layout based on the required room with additional for the daily purposes of students and lecturer to meet the needs of daily users. This project focuses on planning and designing an infrastructure that supports the faculty's growth objectives and technological requirements, ensuring it remains both efficient and secure. The success and grade of the project will depend on the team's ability to collaborate effectively and produce a well-thought-out, high-quality design.

4.1.2 Floor plan for Ground Floor

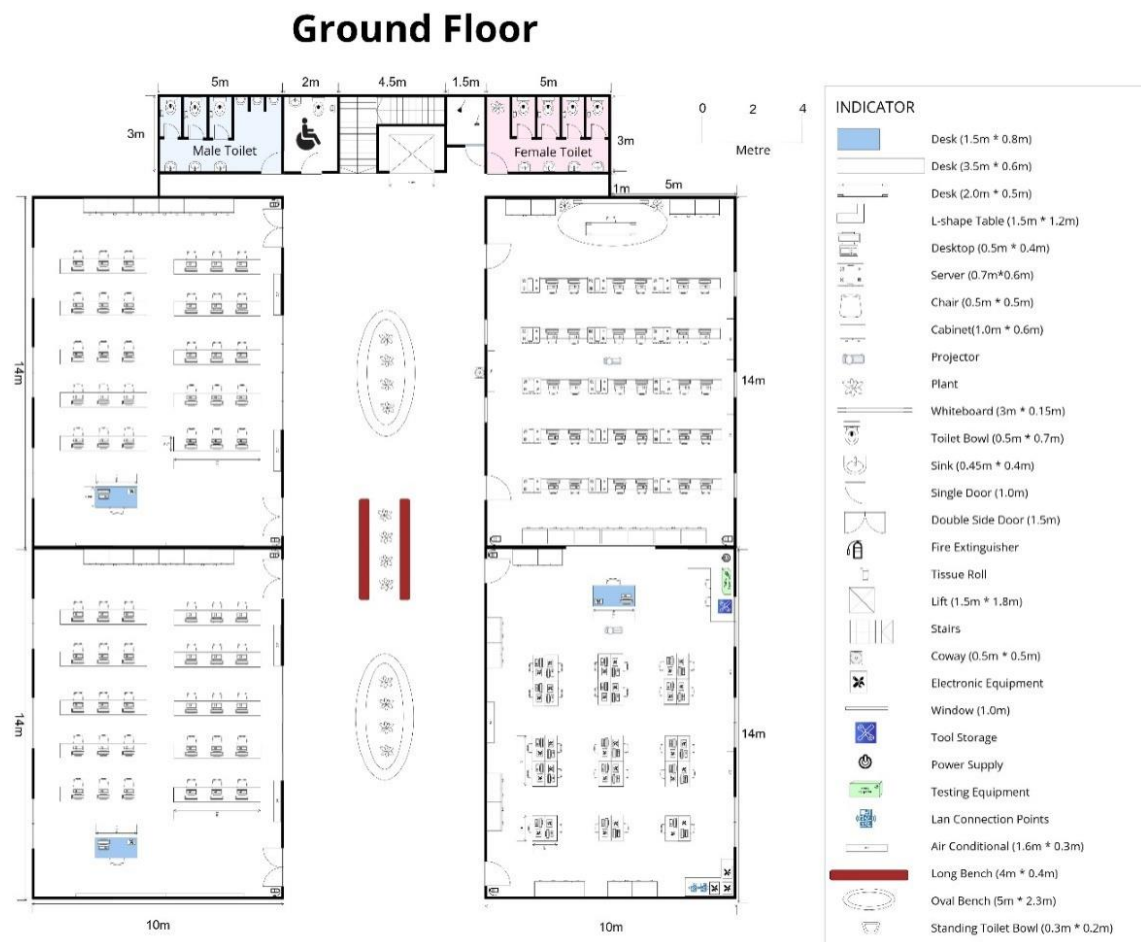


Figure 4.1.2.1 Floor plan for ground floor

4.1.3 Floor plan for First Floor

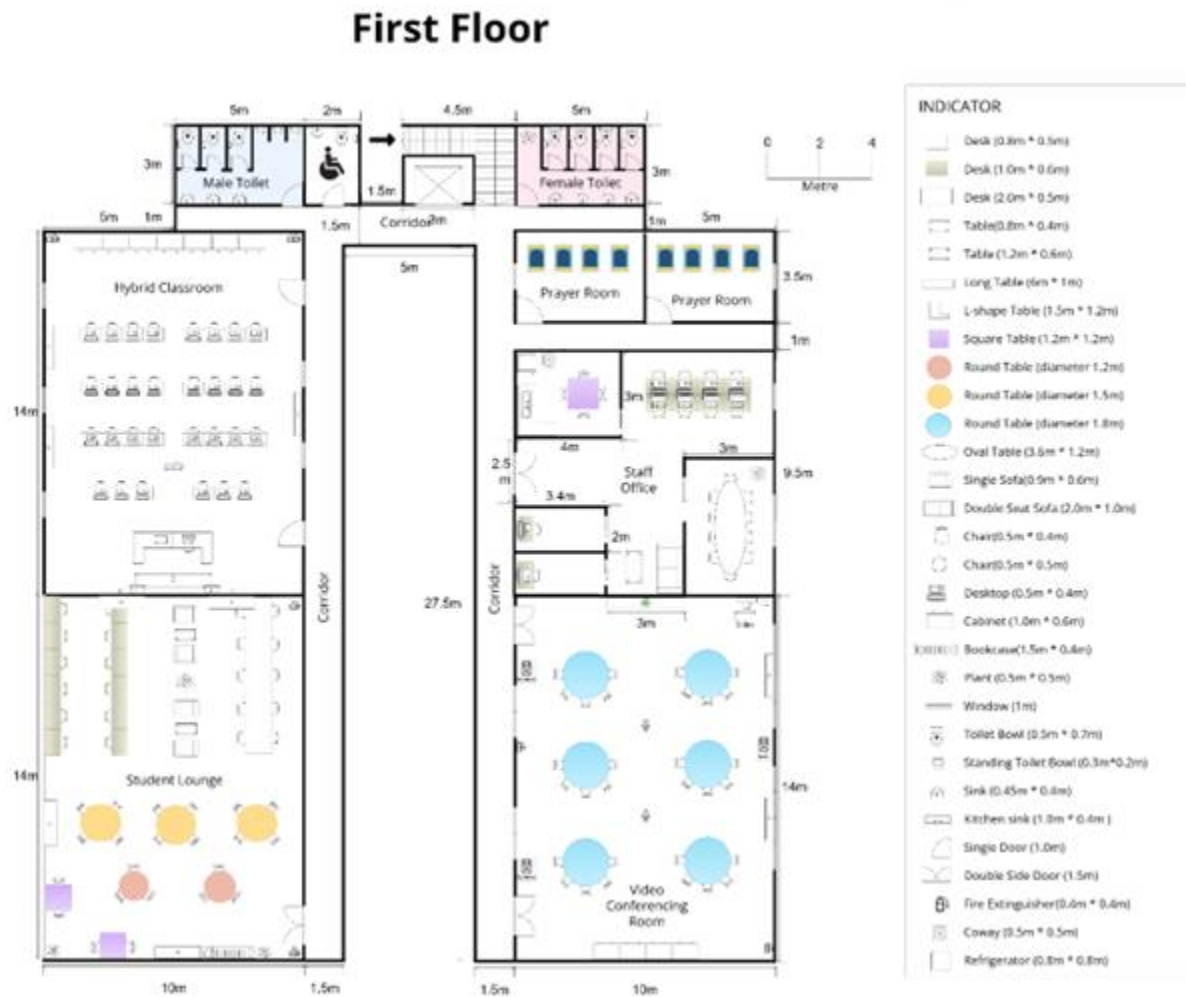


Figure 4.1.3.1 Floor plan for first floor

4.1.4 Brief Explanation for The Layout

We are establishing four laboratories on the ground floor which are two general laboratories, a Cisco Network Lab, and an Embedded Lab. The two general labs are positioned on the left side of the building, while the Cisco Network Lab and the Embedded Lab are located on the right side. Centrally situated is a lift, surrounded by stairs. At the back of the left wing, there is a male restroom, along with an accessible restroom next to the stairs. Conversely, the female restroom is located on the opposite side of the stairs. The center of the building features greenery and benches for students to utilize.

For the first floor, we have designated spaces for the Student Lounge, Staff Office, Hybrid Classroom, and Video Conferencing Room. The Student Lounge is strategically positioned on this level due to its quieter environment, minimizing people passing by and creating a more conducive atmosphere for students. As illustrated, the Student Lounge and Hybrid Classroom are located on the left side, while the Staff Office and Video Conferencing Room are situated on the right side of the building. Additionally, we have allocated two prayer rooms beside the staff room for the male and female Muslims separately. Finally, the design and placement of both the female and male restrooms mirror those on the ground floor.

4.1.5 Enhanced Floorplan by using Packet Tracer

Ground floor

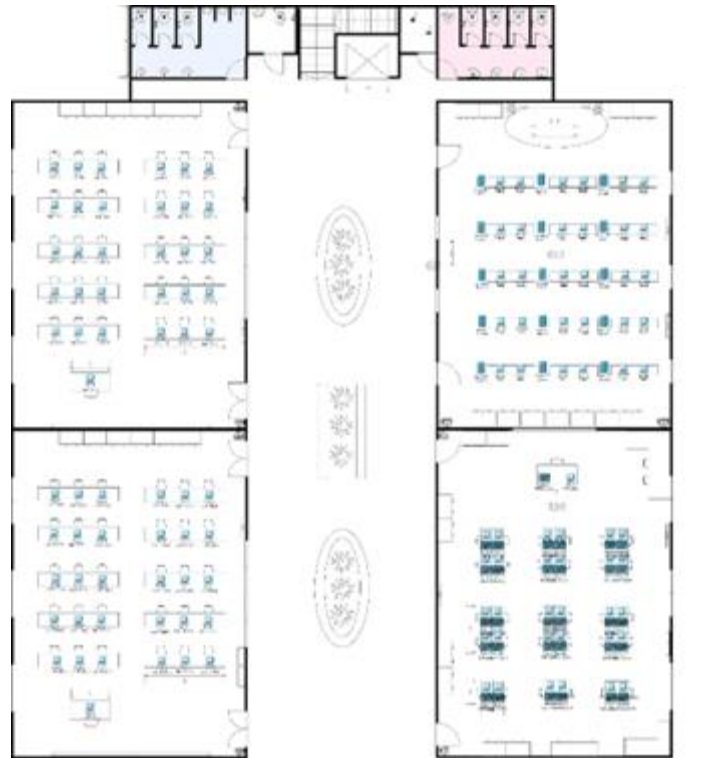


Figure 4.1.5.1 Enhanced floorplan by using Packet Tracer for ground floor

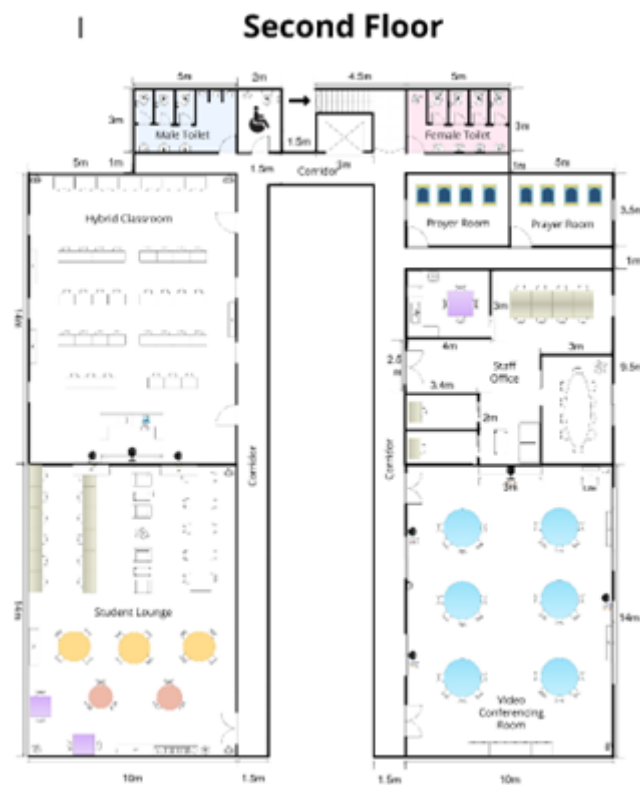


Figure 4.1.5.2 Enhanced floorplan by using Packet Tracer for first floor

4.1.6 Task 1 Reflection

This project was an insightful and valuable learning experience, as it required our team to collaborate, research and apply our technical knowledge to design a two-story building with an optimized network infrastructure. The process of planning and implementing the floor plan using tools like Visual Paradigm and Packet Tracer helped us enhance our technical and problem-solving skills. From the initial discussions to the finalization of our design, we encountered various challenges that required critical thinking and teamwork to overcome.

One of the key takeaways from this project was the importance of effective communication and teamwork. Through multiple meetings, we discussed ideas, allocated tasks and made collective decisions to ensure that every aspect of the design was well thought out. Our group successfully divided responsibilities based on individual strengths, which improved efficiency and productivity. However, coordinating schedules and maintaining consistency in design were some of the challenges we had to manage. Regular feedback and constructive discussions helped us refine our design and improve our approach.

Additionally, using Packet Tracer to enhance our floor plan provided hands-on experience in designing and simulating network layouts. This process not only strengthened our understanding of network configurations but also emphasized the need for precision and standardization in a well-structured computing lab. Overall, this project has been a rewarding experience that has enhanced both our technical and interpersonal skills. Moving forward, we will continue to apply the lessons learned in future projects, ensuring better planning, coordination and execution.

4.2 Task 2

4.2.1 Question

1. What are the current pain points with the existing networking setup?

There are frequent connectivity issues during peak usage times, especially in high-density areas like lecture halls and labs and also some areas with lower or even no connection. We also experience slower speeds and intermittent dropouts during video conferences and virtual meetings, which can disrupt communication and collaboration.

2. What types of network devices will be required for the Cisco Network lab, video conferencing room and embedded lab?

The Cisco Network Lab will require appropriate networking equipment, such as routers, switches, and firewalls, to facilitate the teaching of Cisco-related concepts.[6] Similarly, the video conferencing room will necessitate high-quality audiovisual components, including cameras, microphones, speakers, and a reliable, high-bandwidth internet connection, to enable effective remote communication and collaboration. Lastly, the embedded lab will need a range of specialized hardware, including logic analyzer and Raspberry Pi kits, microcontrollers, digital oscilloscopes, and sensor modules, in addition to relevant software tools like MATLAB, LabVIEW, and IoT cloud platforms, to support the hands-on exploration and development of embedded systems.[7, 8]

3. What are the essential network and hardware requirements for an effective and high quality video conferencing room?

The hardware requirement for an effective and high-quality video conferencing room is having a high-resolution webcam. It is essential for clear video communication, as it captures subtle details like facial expressions and body language, enhancing the authenticity of virtual interactions. The sufficient network bandwidth is also critical to achieve the requirement. For example, a moderate-quality 720p video at 15 frames per second requires about 1 Mbps, while a high-quality 4K video at 30 frames per second

demands approximately 4 Mbps. The amount of internet strength needed depends on how clear you want the video to be. Additionally, when multiple participants join a video call, higher bandwidth is necessary to maintain smooth connectivity. Even with strong Wi-Fi coverage, simultaneous usage can strain the network, the network must be capable of handling concurrent usage to maintain uninterrupted and high-quality video conferencing.

4. What network services will be offered to support teaching and learning activities?

Provide virtual learning platforms hosted on local servers or the cloud, secure access to digital libraries, research databases, and online courses. Implement streaming services for live lectures and virtual labs to enhance interactive learning. For example, Cisco Education which securely connects educators, students, and administrators with transformative technologies that power inclusive learning for all[15].

5. How can the building's wireless network support high-density areas like the student lounge and labs?

To effectively support high-density areas, such as the student lounge and labs, the building's wireless network should utilize enterprise-grade access points equipped with multi-user, multiple-input, multiple-output technology and band steering capabilities. Strategically positioning the access points can optimize wireless coverage. Additionally, implementing Wi-Fi 6 access points can facilitate simultaneous connections for numerous devices in each floor.[10] Further, segregating the wireless network into separate service set identifiers for students, staff, and guests, while imposing bandwidth restrictions for non-critical users, can help manage network usage efficiently.

6. What type of network architecture (e.g., star, mesh, bus, or hybrid) would best suit the faculty's needs, considering factors like scalability, reliability, performance, budget, and ease of maintenance?

A hybrid network architecture that combines star and mesh topology is the most suitable design for the new faculty. On the first floor, star topology will be used to connect all devices such as printers, desktops and wireless access points to a central switch to ensure the effectiveness and localized connectivity. Otherwise, mesh topology will be used to link Floor 2's switches to Floor 1's main switch using a high-speed backbone such as fiber-optic or high-capacity Ethernet. This design balances scalability, performance, and cost, enabling easy additions of devices and robust communication between floors.

7. What is the best method of connecting devices to the network (e.g., wired, wireless, or a combination of both)?

The best method for connecting devices to the network is a combination of both wired and wireless connection depending on the type of device and usage. For wired networks, devices are physically connected by using cables to a network switch or router.[20] Wired networks are suitable for devices such as printers, desktop as they are able to provide a stable and high-speed connection with low latency. However, wireless connection using electromagnetic waves to transmit data from end-to-end. Devices like laptops, tablets and smartphones are more suitable to use wireless connection as it is more flexible without using physical cable and ease of installation[20]. Otherwise, Wireless Access Points (APs) are strategically placed throughout the building to provide seamless coverage, allowing users to connect to the network with more stability.

8. What kind of wireless network setup is required to provide seamless connectivity across the building?

To provide seamless wireless connectivity across the faculty building, deploy multiple access points (APs) to ensure consistent coverage in all areas. Use wireless repeaters or extenders to eliminate dead zones in distant or obstructed spaces. Incorporate network switches and controllers for centralized management of APs, streamlining configuration, and improving network performance. Additionally, consider using directional or omnidirectional antennas based on the coverage requirements and ensure all devices

are equipped with network interface cards (NICs) for compatibility with the wireless network[12].

9. What type of network cable is the most appropriate for connecting the computers, and what are the reasons for choosing it?

Cat6 Ethernet cables are the most appropriate type of cabling for the network setup. Cat6 cables support speeds up to 1Gbps, which is sufficient for typical educational applications and ideal for labs with the need of high speed connectivity[16]. Additionally, Cat6 cables have a good balance between performance and cost as they are widely available and easy to source. Cat6 cables are compatible with existing networking equipment such as routers and switches that support Gigabit Ethernet, ensuring that the faculty's network infrastructure work efficiently without requiring an upgrade[16].

10. How can network efficiency and low latency be achieved?

Low latency and network efficiency require the use of several techniques. A network design that is well-structured, like a core-distribution-access topology, guarantees effective traffic flow by lowering congestion and the distance that data must travel. Important applications like video conferencing may function without interruption thanks to network segmentation using VLANs, which helps isolate various traffic types.[2] By prioritising time-sensitive traffic, Quality of Service (QoS) implementation guarantees minimal latency for applications with high demand.[1] To maximise speed and reduce delays, high-performance gear is necessary. Examples of this hardware include fiber-optic cabling for inter-floor connections, low-latency access points, and gigabit or 10-gigabit Ethernet switches. The network is further optimised by traffic prioritization including load balancing, traffic shaping, and redundancy, which distribute traffic uniformly and offer backup routes to avoid congestion.[1]

11. What bandwidth and internet connectivity are required to support the faculty's operations?

A minimum bandwidth of 1 Gbps is recommended, with provisions to scale to 10 Gbps in the future[11]. Bandwidth is critical to the success of hardware initiatives that attempt to provide ubiquitous access to learning, such as Bring Your Own Device (BYOD) and 1:1 laptop and tablet programs. The amount of bandwidth needed is obviously driven by the types of networked activities in which the user is engaged such as downloading content, online professional developments and online testing.

12. How can we ensure that the wireless network covers the entire faculty building effectively?

To ensure that the coverage wireless network is able to support the area of the faculty, a thorough evaluation of signal strength across all areas, including floors, common spaces and specific rooms is essential[19]. This evaluation involves determining the quantity of current Access Points (APs) and their effective range to establish the extent of network coverage [19]. Detect and scrutinize dead zones, areas with weak or non-existent signal strength [19], to pinpoint areas in need of improved coverage. Considering the characteristics of the wireless network, including factors like Wi-Fi channels, network authentication, and security protocols, make necessary adjustments to optimize coverage and ensure a dependable connection. Based on the findings and analysis, identify suitable new locations for the placement of APs to enhance coverage and address the previously identified dead zones. Carefully position the new APs to ensure high-quality speed and coverage precisely where it's needed.

13. How should the building's network be secured against external and internal threats?

The Cybersecurity and Infrastructure Security Agency (CISA) encourages users and network administrators to implement the segment and segregate networks and functions to better secure their network infrastructure. Proper network segmentation is an effective security mechanism to prevent an intruder from propagating exploits or laterally moving around an internal network. A securely segregated network can contain malicious occurrences, reducing the impact from intruders in the event that they have gained a foothold somewhere inside the network[13]. Moreover, implementing a

Virtual Private Network (VPN) is essential[14]. A VPN encrypts data transmissions and masks users' IP addresses, providing secure access to the network, especially for remote users or those connecting via public Wi-Fi. This helps protect sensitive information from unauthorized access and cyber threats.

14. What disaster recovery plan is needed for network downtime?

Planning for disaster recovery is essential to ensure an organization can maintain operations during unexpected events such as natural disasters, cyberattacks, or other emergencies. Regular data backups can ensure the important files and settings are safe and can be recovered easily. Implementing a backup network system, like additional routers and internet connections can keep the network running if the main system fails[17]. The real-time monitoring tools play a crucial role in detecting issues early by sending alerts, so issues can be addressed quickly[17]. Furthermore, Uninterruptible Power Supplies(UPS) can be used to protect devices from power outages as it can give more time to fix the issue without losing connectivity[18]. These measures collectively strengthen the organization's resilience and minimize the impact of disruptions.

15. What monitoring tools are recommended for detecting and managing network issues?

Several monitoring technologies can be used to provide real-time insights, identify bottlenecks, and guarantee optimal performance in order to detect and address network issues efficiently. With comprehensive data on bandwidth utilisation, latency, and device condition, SolarWinds Network Performance Monitor is a well-liked tool for keeping an eye on the functionality and health of network devices. It can detect and solve network issues before they cause downtime.[3] With customisable alerts to inform administrators of any problems, PRTG Network Monitor provides a complete solution for tracking network traffic, server performance, and device health. Another open-source tool for network and infrastructure monitoring is Nagios, which can measure a number of metrics like response times, availability, and uptime.[3] In order to solve problems like network congestion or security vulnerabilities, network administrators

can collect and examine packets using Wireshark, which allow us to capture and analyse network traffic. [4]

4.2.2 Feasibility Assessment

Feasibility assessments are critical in determining whether a project is realistic and worth pursuing. After the Question and answer part, we can now evaluate and discuss the feasibility of our project. These are the four feasibility we are going to determine and evaluate:

1. Technical Feasibility

Based on the information gathered, the project appears to be technically feasible. The proposed network architecture features a fiber-optic backbone and Cat6 cabling to support high-speed connectivity and scalability for faculty. We also select enterprise-grade access points with Wi-Fi 6 to ensure coverage in high-density areas like labs and lounges to meet future demands. The integration of devices such as Raspberry Pis, microcontrollers, and Cisco equipment meets the labs' technical and educational needs. Lastly, the adoption of modern technologies like VLANs for traffic segmentation, Quality of Service (QoS) for latency-sensitive applications, and monitoring tools like SolarWinds and Nagios enhance network reliability and performance.

2. Economic Feasibility

It is essential to evaluate the costs and benefits of the proposed investment to assess the economic feasibility of the project. The initial costs include purchasing equipment, software installation and software license fees while operational costs include maintenance, IT management staffing and power consumption. However, the project offers significant economic advantages, such as reduced downtime, increase productivity, and long-term savings from scalable and future-proof technology like Wi-Fi 6 and Cat6 Ethernet cabling. A cost-benefit analysis which compare the costs with

the benefits it will provide will help in ensuring the project stays within budget while achieving the faculty's goals effectively.

3. Operational Feasibility

The operational feasibility of the project appears promising. By using a hybrid network architecture which combines star and mesh topologies helps to ensure the connectivity and seamless communication between floors. Otherwise, high-performance hardware such as managed switches, routers and Wi-Fi 6 access points support efficient wired and wireless connections. Moreover, for solving traffic segregation, we have included VLANs and Quality of Service(QoS) to keep the lower latency while using video conferencing and hybrid labs. Monitoring tools like SolarWinds enable proactive issue management. While strong security standards protect data, but disaster recovery methods, such as backups and redundant systems can improve resilience.

4. Security Consideration

To secure the faculty building's network, segmentation and segregation of networks are essential to contain threats and limit intruder movement within the network. Implementing Virtual Private Networks (VPNs) ensures encrypted communication for remote access and public wi-Fi usage. A robust disaster recovery plan is necessary such as redundant systems and regular backups, making the network secure and reliable for educational operations.

4.2.3 Task 2 Reflection

This group project has been a valuable experience that allowed us to deepen our understanding of network design and infrastructure planning. From the initial brainstorming sessions to the final feasibility assessment, we encountered various challenges that required critical thinking, collaboration and technical research. The project not only tested our knowledge of networking concepts but also improved our ability to analyse real-world problems and propose effective solutions.

One of the major lessons learned during this project was the importance of teamwork and task delegation. Each member was assigned specific responsibilities, such as researching network architecture, selecting suitable hardware and designing the floor plan. By distributing tasks effectively, we managed to work efficiently while ensuring that every aspect of the network design was covered. Regular discussions and meetings helped us refine our ideas and ensure that all components of the project aligned with the faculty's needs.

Additionally, we faced challenges in designing a network that would support high-density areas such as the student lounge and labs while ensuring seamless connectivity and security. Addressing issues like bandwidth allocation, network segmentation and disaster recovery planning helped us gain a deeper appreciation for the complexity of network communications in an educational setting. Implementing modern technologies like VLANs, Quality of Service (QoS) and enterprise-grade Wi-Fi solutions provided us with insights into real-world networking best practices.

Overall, this project has been an enriching experience that strengthened both our technical and soft skills. It emphasized the importance of structured problem-solving, effective communication and adaptability in tackling complex projects. Moving forward, the knowledge gained from this project will be instrumental in future endeavours, whether in academic research or professional networking roles.

4.3 Task 3 Network Device Selection

4.3.1 Devices Introduction

In designing an efficient network infrastructure for this project, it is essential to incorporate the appropriate communication devices that ensure seamless connectivity, high performance, and reliability. This section highlights the key network communication devices required for the project, along with their roles and specifications to meet the project's requirements.

1. Router

A router is a networking device that enables communication between multiple computer networks by directing data packets to their intended destinations. It functions as a central coordinator, analysing incoming data packets and deciding the most efficient route to forward them using Internet Protocol (IP) addresses to ensure data packets are delivered accurately. Routers operate by using routing tables or algorithms, which map out the available paths for data transmission. Routers can link local devices within a private network (like in homes or offices) and facilitate connections to external networks, such as the internet. Advanced routers even support features like Quality of Service (QoS), network security through firewalls, and traffic prioritization to ensure efficient performance. Routers are vital components in networking infrastructure to enable data exchange across interconnected systems.

2. Switches

A network switch is a crucial device in computer networking that connects multiple devices and manages data traffic within a network. It directs data packets to their intended destinations using MAC(Media Access Control) addresses, ensuring that only the intended device receives data. Switch can be hardware devices that manage physical networks or software-based virtual devices. It features multiple Ethernet ports which used to connect different devices such as computers, printers, and other network equipment.

3. Patch Panels

Patch panels are device that organizes and manage connections in a network. It serves as a central point for connecting network cables in a structured and efficient manner. It

provides a way to keep large numbers of cables organized, enabling flexible connectivity into network hardware. Patch panels are commonly used in network environments, such as data centres, labs and offices, to simplify cable management and facilitate troubleshooting. There are 3 types of patch panels which is ethernet patch panels, fiber optic patch panels and audio/video patch panels. Otherwise, using patch panels brings benefits like improving cable management, easier troubleshooting, flexibility, durability and scalability.

4. Server

A server is a critical device in any network infrastructure, designed to manage, store, and process data while handling applications and network resources for multiple users simultaneously. It acts as the central hub for data storage, application hosting, and resource sharing, ensuring seamless operation across the network. Servers support centralized file management, learning management systems, databases, and user authentication, providing a secure and scalable platform for academic and administrative functions.

5. Wifi Access Point

Wifi Access Point is a networking device that allows wireless devices to connect to a wired network. It acts as a bridge, facilitating communication between wireless devices such as smartphones, laptop, IoT devices with network infrastructure. Access points are needed in environments where many users need high-speed connectivity. The advantages of using Wifi Access Point is high scalability, high performance and high flexibility.

6. Projector

Projector is an optical device used to display images or videos onto a screen or flat surface, either via wired or wireless connections. It is essential for effective communication which allows instructors to present lectures and demonstrations to large audience. Additionally, projector may support advanced features like build-in speakers for audio output and high-resolution visuals which enhancing the learning experience.

7. Desktop

Desktops are essential workstations used for individual tasks such as programming, document creation, simulation, and multimedia editing. Desktops are deployed in labs, classrooms, and offices to facilitate student projects, software development, and administrative tasks. Their performance, reliability, and compatibility with academic software make them indispensable for daily operations in education and research environments.

8. Smartboard

A smartboard is a self-contained interactive whiteboard with a built-in operating system. It allows users to write, draw, and manipulate objects directly on its touch-sensitive screen. Smartboard can display digital content, access the internet, and utilize various applications without needing an external device. It supports real-time collaboration which enhancing productivity and engagement in presentations and meetings.

9. Cat6 Ethernet Cable

A Cat6 Ethernet cable is a high-performance networking cable designed to transmit data at high speeds with minimal interference. It supports data transfer rates of up to 1 Gbps over distances of 100 meters and can handle speeds up to 10 Gbps for shorter distances (up to 55 meters). Cat6 cables are constructed with tightly twisted pairs of copper wires. This design minimizes crosstalk and electromagnetic interference (EMI), ensuring reliable and fast data transmission, even in environments with high electrical noise. These cables are widely used in networking applications, such as connecting computers, routers, switches, and servers in data centers. In summary, Cat6 Ethernet cables are a cost-effective, high-speed solution for wired networks, offering durability and excellent performance for both current and future networking needs.

10. Fiber Optic Cables

Fiber optic cables are high-speed communication mediums that transmit data using light signals, ensuring minimal signal loss over long distances. These cables are crucial for providing the high-speed connectivity required in a modern faculty building, serving as the backbone for data transmission between servers, switches, and routers. With their

superior bandwidth, reliability, and scalability, fiber optic cables support seamless communication and future-proof the network infrastructure.

4.3.2 Table of Devices Needed

Devices	Description	Price per Unit	Quantity	Total price (RM)
Routers  Figure 4.3.1.1 Routers	Ubiquiti EdgeRouter 12 – System memory: – 1 GB DDR3 RAM – On-board flash storage: 4 GB eMMC, 8 MB SPI NOR – Networking interface: (10) GbE RJ45 ports, (2) 1 Gbps SFP ports – Management interface: (1) RJ45 serial port – Power method: External AC/DC power adapter, 24W, 24V, 1A (Included) , 24V Passive PoE – Power supply: External AC/DC adapter – Passive PoE Voltage range Max. wattage per port: DC/PoE passthrough 20W (24V) – Supported voltage range: 9-30V DC	1,269	4	5076

	– Max. power consumption: 20W (Excluding PoE output)			
Switches  Figure 4.3.1.2 Switches	S3400-24T4FP - Ports: 24x 10/100/1000BASE-T RJ45 4x 1G RJ45/SFP Combo - Fans: 3 Built-in Fans - PoE Budget: 370W - Material: Metal - Features: 1) Virtual deployment of dual switches offers flexible and high-performance network for serving all scenarios on campus 2) Support web management, firmware upgrade and configuration file upload/download via TFTP/HTTP/FTP server, and firmware auto upgrade Remote Monitoring (RMON).	2033	13	26429
Patch Panels  Figure 4.3.1.3 Patch panels	BELNET CAT6 24-Port Patch Panel - Type: CAT6 24-Port Patch Panel - Form Factor: 1U rack - Network Support: Gigabit Network (Transmission up	378	13	4914

	to 250MHz, supports speeds of 1G – 10G Mbps) - Performance Standard: Tested according to ISO/IEC 11801 CLASS E PL2. - Front Panel: Include label field for easy identification and have RJ45 front sockets with a smart shutter - Rear Side: Equipped with a cable management metal bar for organizing and securing cables - IDC Termination: Provide Termination Caps, compatible with 568A & 568B color codes, supports cable wire diameters from AWG23 to AWG26 - Package includes Screws set (x4), Cable ties (x24) - Weight: 0.9kg - Dimensions: 52x9.5x6cm			
Servers  Figure 4.3.1.4 Server	Dell PowerEdge R750 Rack Server - Form factor: 2U rack - Processors: Up to 2 x 3rd Generation Intel® Xeon® Scalable processors; up to 40 cores per processor - Memory: Up to 3200MT/s	30,895	4	123,580

	- Storage: Up to 12 x 3.5-inch SAS/SATA (HDD/SSD), max 192TB - Networking: 2 x 1 GbE LOM and 1 x OCP 3.0 (x8 PCIe lanes) (optional) - NVMe Support: Up to 24x NVMe (HDD/SSD) - Management: Embedded iDRAC9			
Wifi Access Point  Figure 4.3.1.5 Wifi access point	Cisco Catalyst 9130AX series - Wireless Communication Standard: 802.11ax - Antenna Type: Internal antenna - Radio Chains and Streams: 4x4 4MIMO - Frequency Band Class: Dual-Band - Frequency: 2.4GHz and 5GHz - Speed: 5.38 Gbit/s - Bluetooth Standard: Bluetooth 5 - Client Capacity: Supports over 1000 clients - Security: WPA3, Cisco TrustSec and identity-based access	3300	10	33000
Projector 	Epson PowerLite X49 - Brightness: 3600 lumens - Connectivity: HDMI,USB,VGA,Wireless	1799	5	8995

Figure 4.3.1.6 Projector	- Resolution: XGA (1024*768) - Contrast: 22000:1 - Lamp Life: Up to 15000 hours - Audio: Built-in 2W speaker - Projected distance: 1.189-13.106 m - Features: Technology: Digital Light Processing (DLP), sharper images, compact size			
Desktop  Figure 4.3.1.7 Desktop	HP Elite Tower 800 G9 Desktop - Processor: Intel® Core™ i5-13500 - Graphics: Intel UHD Graphics 770 - Ports: Provides a SuperSpeed USB Type-C port with 20Gbps - Memory and Storage: 512GB SSD	4652	164	762928
SmartBoard  Figure 4.3.1.8 Smartboard	Huawei IdeaHub S2 - Operating System: HarmonyOS - Connectivity: WiFi-6, Bluetooth, USB, HDMI - RAM: 8GB - Storage: 128GB - Resolution: 4K Ultra HD (3840*2160)\ - Display Size: 65", 75", 86"	15000	1	15000

	- Camera: 4K Video Camera - AI features: Voice tracking, auto- framing and acoustic baffle - Latency: Ultra-low 16ms			
Uninterrupted Power supply (UPS) 	CyberPower UPS OL10KRT - Capacity: 10kW - Topology: Double-Conversion - Runtime: expandable with additional battery packs - Remote Management: PowerPanel software enables to monitor and manage UPS remotely - Ports: USB, Serial, Relay, EPO - Energy saving: ECO mode efficiency up to 97% - Fast Charge Technology: Yes	36000	1	36000
Cat6 Ethernet Cable 	Mediabridge Ethernet Patch Cable (Cat6) - Brand: Mediabridge - Connector Type: RJ45 - Cable Type: Ethernet - Compatible devices: computer, router, modem, television - Special feature: high speed, data transfer, low voltage	80/ 15m	1500m	8000

Figure 4.3.1.9
Uninterrupted power supply (UPS)

Figure
4.3.1.10 Cat6 ethernet cable

	– High Speed: Ultra Fast Throughput of 10 Gigabit per Second at 500 MHz – Use: Easily handles the most demanding home use such as Gaming, High-Definition Video Streaming, Cloud Computing etc. – Server applications: 10 gigabit throughput at up to 250 MHz guarantees high-speed data transfer for server applications. Suitable for 10BASE-T, 100BASE-TX (Fast Ethernet), 1000BASE-T/1000BASE-TX (Gigabit Ethernet) and 10GBASE-T (10-Gigabit Ethernet) applications.			
Connector  Figure 4.3.1.11 Connector	Cat6 RJ45 Modular Plug (8P8C) - Ethernet Type: Rj45 Connector - LAN category: Cat6, Cat6a - Contact Plating: Gold - Mounting Type: Cable Mount	0.38	500	190
Fiber Optic Cables	SC APC Patch Cable, OS2 - Fiber count: 1 fiber - Wavelength: 1310/1550nm	14/ 1m	50	700

 Figure 4.3.1.12 Fiber optic cables	- Insertion Loss: $\leq 0.3\text{dB}$ - Return Loss: $\geq 60\text{dB}$			
			TOTAL	1024812

4.3.3 Major differences between the same devices from different brands

Routers

Here's a detailed comparison table between the **Ubiquiti EdgeRouter 12** and the **TP-Link Omada ER7206** routers, which are suitable for university faculty use.

Feature	Ubiquiti EdgeRouter 12	TP-Link Omada ER7206
Price	MYR 2,200 - MYR 2,800	MYR 1,500 - MYR 2,000
Ports	12 Gigabit Ethernet ports (1 SFP port for fiber)	6 Gigabit Ethernet ports
Processor	1.7 GHz Dual-Core Processor	1.7 GHz Quad-Core Processor
Wi-Fi Support	No built-in Wi-Fi support, requires external APs	No built-in Wi-Fi support, requires external APs
Routing Protocols	Supports static routes, OSPF, BGP, and more	Supports static routing, OSPF, RIP, and more
Management	Web-based interface (Unifi Controller can be used separately)	Centralized management via Omada SDN Controller (Cloud-based)
VLAN Support	Yes, supports advanced VLAN configurations	Yes, supports VLAN configuration with dynamic VLAN tagging
Firewall	Advanced firewall features, with DPI (Deep Packet Inspection)	Advanced firewall features, including DoS protection

Security Features	IP Filtering, DoS Protection, Stateful Packet Inspection	IP Filtering, DoS Protection, VPN passthrough, SPI
VPN Support	Supports OpenVPN, L2TP/IPsec, PPTP, and more	Supports OpenVPN, L2TP, PPTP, and IPsec VPN
Power over Ethernet (PoE)	Yes, on some ports (PoE+ on 2 ports)	Yes, PoE+ on 4 ports
Performance (Routing Throughput)	Up to 3.4 Gbps routing throughput	Up to 1.7 Gbps routing throughput
Suitable for	Larger setups with high network traffic needs , including more advanced routing	SMBs and faculties with moderate network traffic demands

According to the comparison above, since our faculty plan to establish multiple labs, including a Cisco networking lab and a video conferencing room, indicates the need for large and complex setups. We require routers with additional ports and greater routing throughput, making the Ubiquiti EdgeRouter 12 an excellent option. It features six more Gigabit Ethernet ports than the TP-Link Omada ER7206 and provides a higher routing throughput of 3.4 Gbps, making it ideal for more extensive network configurations. Additionally, it offers advanced routing capabilities like OSPF and BGP, which are perfect for intricate network environments, along with limited PoE support on specific ports for external access points.

Switches

We have chosen **S3400-24T4FP** switches to use in this project. The major difference between S3400-24T4FP with other different brands we can compare is **NetGear ProSAFE GS728TP**.

Specification	S3400-24T4FP	NetGear ProSAFE GS728TP
Packet Buffer Size	500KB	512KB
Ports	24 x 1Gbe + 4 Combo Uplinks	24 x 1Gbe + 4 Combo Uplinks
Cooling	3 Built-in Fans	Fanless
PoE Budget	370W	380W

Power Consumption	400W	300W
Ability to Handle Traffic	Yes with advanced L2+ Features	Yes with advanced L2+ Features
Power Efficiency	High	High
Material	Metals	Metal
Price	RM 2033	RM 2900

Based on the comparison above, both are robust options for network environments, but the S3400-24T4FP offers a better balance of performance and affordability. Although the NetGear model has a slightly higher PoE budget of 380W and lower power consumption of 300W compared to S3400's 370W and 400W, the S3400 have 3 built-in fans for effective cooling, ensuring reliable operation under heavy loads, whereas the fanless design of NetGear may lead to potential overheating in high-demand scenarios. Both switches feature same ports number and advanced layer 2+ traffic management capabilities, but the S3400 achieves this at much lower price of RM2033, significantly undercutting the NetGear's RM2900. This makes the S3400-24T4FP the more cost-effective and practical choice for most network setup.

Patch Panels

For patch panels in this project, we have chosen BELNET CAT6 24-Port Patch Panel. The major difference between **BELNET CAT6 24-Port Patch Panel** with other different brands we can compare is **Mini-Com® Patch Panel by Panduit**.

Specification	BELNET CAT6 24-Port Patch Panel	Mini-Com® Patch Panel by Panduit
Category	CAT6	CAT6
Ports	24	24
Size (Rack Units)	1U	1U
Design	Simple, functional front with label fields for port IDs.	Modular design with label fields for easy identification.
Cable Management	Includes a metal bar for cable management at the rear.	Modular design allows flexibility with cable routing.

Shutter Protection	Dust-protective shutters on RJ45 sockets.	No shutters
Color-Coded Labels	568A & 568B for installation ease.	Provides clear labelling options for modular configurations.
Wire Support	Accepts AWG23 to AWG26 cables, solid or standard	Compatible with various cable types based on modular jacks
Tool Compatibility	Suitable for Krone and 110 punch-down tools	Compatible with modular components
Construction Material	Metal chassis for durability	Robust design with high-quality materials
Testing	Fluke-tested for performance assurance	Meets or exceeds industry standards like TIA/EIA
Price	RM 378	RM 627.13

The major differences between devices from different brands lie in build quality, features, performance, cost and support. For our faculty, BELNET is more suitable as it focusses on affordability and basic functionality, suitable for smaller-scale setups. Although Panduit offer advanced functionality, better materials and robust support, but its price is much higher and will exceed our budget. In conclusion, BELNET will be our choice as it reaches our requirement and budget.

Server

For the servers used in this project, we have chosen **Dell PowerEdge R750 Rack Server**. The major difference between **Dell PowerEdge R750 Rack Server** with other different brands we can compare is **Lenovo ThinkSystem SR650**.

Specification	Dell PowerEdge R750 Rack Server	Lenovo ThinkSystem SR650
Form factor	2U rack	2U rack
Processors	Up to 2 x 3rd Generation Intel® Xeon® Scalable	Up to 2x 3 rd Gen Intel® Xeon® Scalable processors

	processors; up to 40 cores per processor	
Memory	Up to 3200 MT/s	Up to 32x 128GB TruDDR4 3200MHz memory
Storage	Up to 12 x 3.5-inch SAS/SATA (HDD/SSD) max 192 TB	Up to 32x NVMe drives
Networking	2 x 1 GbE LOM and 1 x OCP 3.0 (x8 PCIe lanes) (optional)	LOM adapter installed in the OCP 3.0 slot; PCIe adapters
NVMe support	Up to 24x NVMe (HDD/SSD)	Up to 32x NVMe drives
Management	Embedded iDRAC9	Lenovo XClarity Controller
Price	RM30,895	RM16,945

Based on the table above, the Dell PowerEdge R750 Rack Server is the more suitable option for the faculty due to its superior performance and advanced capabilities. With support for up to 40 cores per processor and versatile storage options, including SAS/SATA and NVMe drives, it is ideal for resource-intensive tasks such as simulation, virtual labs and research. The integrated iDRAC9 management system ensures efficient remote monitoring and troubleshooting, enhancing operational reliability. Additionally, its flexible networking options via PCIe lanes provide scalable connectivity solution, making the Dell PowerEdge R750 a powerful and future-ready choice for a demanding academic environment. However, it comes at a higher cost compared to its counterpart.

Wi-Fi Access Point

For the Wi-Fi Access Point used in this project, we have chosen Cisco Catalyst 9130AX Series. The major difference between **Cisco Catalyst 9130AX Series** with other different brands we can compare is **Aruba 515**.

Specification	Cisco Catalyst 9130AX Series	Aruba 515
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Wi-Fi Standard	Wi-Fi 6 (802.11ax)	Wi-Fi 6 (802.11ax)
Maximum Data Rate	Up to 5.38 Gbps	Up to 4.8Gbps
Radio Chains and Streams	4x4:4 MU-MIMO	4x4:4 MU-MIMO
Antenna Type	Internal and external antenna options available	Internal antennas
Frequency Bands	2.4GHz and 5GHz Dual band	2.4GHz and 5GHz Dual band
Power over Ethernet (PoE)	PoE+, UPOE and UPOE+ support	Supports PoE (802.3af/802.3at)
Client Capacity	Supports over 1000 clients	Supports over 500 clients
Software Features	Cisco DNA support, CleanAir Pro, AI-enhanced analytics	ArubaOS with AI-powered RF optimization
Security	WPA3, Cisco TrustSec and identity-based access	WPA3, role-based access policies
Management Options	Cisco DNA Center, web GUI, CLI and third-party APIs	Aruba Central, CLI and local management
Environmental Suitability	Indoor and outdoor models available	Indoor model only
Price	RM3300	RM2899.99

The major differences between the same devices from different brands often lie in performance, features, pricing and ecosystem integration. For example, Cisco provide advanced analytics, AI-driven optimization and higher client capacity while Aruba offers cost-effective solutions with simplified management and robust performance. In these two choices, Cisco is more meets our requirement for using in our faculty building although it is more expensive than Aruba. Therefore, Cisco is the better choice for us in this project.

Projector

For the projector used in this project, we have chosen **Viewsonic PA503X XGA** . The major difference between **Viewsonic PA503X XGA** with other different brands we can compare is **Epson PowerLite X49**.

Feature	Viewsonic PA503X XGA	Epson PowerLite X49
Brightness	3600 lumens	3600 lumens
Connectivity	HDMI, USB, VGA, Wireless	HDMI, USB, VGA, Wireless
Resolution	XGA (1024*768)	XGA (1024*768)
Contrast	22000:1	16000:1
Lamp Life	15000 hours	12000 hours
Audio	2W	5W
Projected Distance	1.189 - 13.106 m	0.89-9.12 m
Features	Technology Digital Light Processing, sharper images and compact size	Moderator function, 3LCD technology, better for large images and energy efficient
Price	RM1799	RM2296

Based on the comparison above, the ViewSonic PA503X XGA s a great choice for the classroom since it has a higher lumens brightness and contrast ratio compare to Epson PowerLite X49, it ensures clear and crisp images even in bright environments. The longer projection distance allows flexible placement within the room. Priced at RM1799, it offers excellent value for money. While the Epson PowerLite offers advanced features like enhanced audio and 3LCD technology, the ViewSonic PA503X XGA is more cost-effective and ideal for classrooms where flexibility and budget are important.

Desktop

For the desktop used in this project, we have chosen **HP Elite Tower 800 G9 Desktop**. The major difference between **HP Elite Tower 800 G9 Desktop** with other different brands we can compare is **Lenovo ThinkStation P3 Tower**.

	HP Elite Tower 800 G9 Desktop	Lenovo ThinkStation P3 Tower
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Operating System	Windows 11 Pro	Windows 11 Pro
Processor	Intel® Core™ i5-13500 (up to 4.8 GHz with Intel® Turbo Boost Technology, 24 MB L3 cache, 14 cores, 20 threads)	13th Generation Intel® Core™ i5-13400 Processor (E-cores up to 3.30 GHz P-cores up to 4.60 GHz)
Ports	1 headphone/microphone combo; 4 SuperSpeed USB Type-A 10Gbps signaling rate; 1 SuperSpeed USB Type-C® 20Gbps signaling rate	USB-C (10Gbps), USB-A (5Gbps and 10Gbps), HDMI 2.1 TMDS, dual DisplayPort™ 1.4, Ethernet (RJ45), headphone/mic combo, microphone input, line-out, optional SD card reader, optional expansion cards, and a power input.
Graphics	Intel® UHD Graphics 770	Integrated Graphics
Memory and Storage	8 GB memory; 512 GB SSD storage	8 GB DDR5-4400MHz (UDIMM) memory; 256 GB SSD M.2 2280 PCIe Gen4 TLC Opal
Price	RM 4652	RM 4079

Based on the comparison above, the **HP Elite Tower 800 G9** is more suitable for the faculty due to its superior performance and features. It is powered by an Intel Core i5-13500 processor with a higher clock speed and larger cache which will make it better equipped for multitasking and demanding tasks. The system includes Intel UHD Graphics 770, offering slightly better integrated graphics performance. It provides double the capacity of Lenovo ThinkStation P3 with 512GB of SSD storage for handling applications and data. Moreover, its SuperSpeed USB Type-C port with a 20Gbps signaling rate ensures faster data transfer for modern peripherals. Although its slightly more expensive, the HP Elite Tower 800 G9 offers better overall value and making it the optimal choice for academic and administrative use.

Smartboard

For the smartboard used in this project, we have chosen **Huawei Ideahub 2**. The major difference between **Huawei Ideahub 2** with other different brands we can compare is **Canon smart interactive board**.

Feature	Mediabridge Ethernet Cable	Ultra Clarity Cables Cat6
Display Size	Available in 65", 75", 86"	Available in 70"
Resolution	4K Ultra HD (2840*2160)	4K
Operation System	HarmonyOS	Android and Windows
Connectivity	Wi-Fi 6, Bluetooth, USB and HDMI	Wi-Fi, Bluetooth and USB
RAM	Up to 8GB	-
Storage	128GB SSD	-
Camera	4K Video Camera	4K Video Camera
Latency	16ms Ultra Low Latency	-
Audio Features	Voice tracking, auto framing and acoustic baffle for superior audio quality	Speaking-tracking and auto-framing
Unique Features	BYOM (Bring Your Own Meeting), Intelligent Whiteboard	OCR (Smart Optical Character Recognition)
Price	RM15000-20000	RM20000-25000

Based on the table above, the Huawei IdeaHub S2 may be the better option due to its advanced features, versatile connectivity options, and budget-friendly price. Its ultra-low writing latency and intelligent whiteboard functions make it perfect for interactive teaching and presentations. Additionally, the high-definition video and superior audio quality enhance virtual lectures and meetings, ensuring seamless and engaging learning experience. Although Canon Smart Interactive Board offers these capabilities, but it has a higher price compared to Huawei IdeaHub S2. Huawei IdeaHub S2's flexibility display sizes also cater to different room sizes, further adding to its suitability for educational environments.

Cat6 Ethernet Cable

For the Cat6 Ethernet Cable used in this project, we have chosen the **Mediabridge Ethernet Patch Cable Cat6**. The major difference between **Mediabridge Ethernet Patch Cable** with other different brands we can compare is the **Ultra Clarity Cables Cat6**.

Feature	Mediabridge Ethernet Cable	Ultra Clarity Cables Cat6
Build Quality	Premium PVC jacket with gold-plated connectors for durability and corrosion resistance.	Durable jacket material with snagless connectors for long-lasting performance.
Cable Length Options	Available in 6 ft, 10 ft, 25 ft, 50 ft, 100 ft, and bulk 1000 ft rolls.	Available in 6 ft, 10 ft, 15 ft, 25 ft, 50 ft, and longer lengths up to 100 ft.
Performance Standard	Meets or exceeds Cat6 performance standards, supporting speeds up to 1 Gbps with a bandwidth of 250 MHz.	Meets Cat6 standards, supports speeds up to 1 Gbps, and offers up to 250 MHz bandwidth.
Connector Type	RJ45 connectors with snagless design for easy plugging/unplugging and secure connections.	RJ45 connectors with gold-plating for reduced signal loss and enhanced connectivity.
Cable Type	UTP (Unshielded Twisted Pair), suitable for most general-purpose networking needs.	UTP (Unshielded Twisted Pair), ideal for home, office, and lab environments.
Color Options	Commonly available in white, black, and blue.	Available in multiple colors like black, blue, gray, and white.
Price	RM 45 - RM 65 for 6 ft, RM 135 - RM 225 for 100 ft. Bulk 1000 ft: RM 365 - RM 550.	RM 45 - RM 90 for 6-15 ft, RM 135 - RM 225 for longer lengths (up to 100 ft).
Ease of Use	Flexible and easy to manage; compatible with Ethernet-enabled devices, including routers, switches, and PCs.	Flexible cable with snagless boots, suitable for all Cat6-compatible devices.
Ideal Use Case	Suitable for general networking in labs, offices, or homes where high-speed, reliable connections are needed.	Good for university labs, home offices, and general networking applications.

Unique Features	Bulk 1000 ft roll is an affordable option for large-scale wiring projects.	Slightly more color options for cable organization and aesthetics.
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Both cables perform to Cat6 standards, making them comparable in speed and bandwidth. However, the Mediabridge offers a slight advantage in pricing for bulk purchases, as 1000 ft rolls are more economical than buying multiple shorter lengths for faculty labs use. Mediabridge is well-suited for large-scale networking projects due to its bulk option, while Ultra Clarity is preferable for smaller setups that require a small range of lengths with different colours. For our project, the Mediabridge Ethernet Cable is the recommended better choice to connect numerous devices or plan extensive wiring across labs and areas, with the bulk roll option being an added benefit.

Fiber Optic Cables

For the Fiber Optic Cable used in this project, we have chosen the **SC APC Patch Cable, OS2**. The major difference between **SC APC Patch Cable, OS2** with other different brands we can compare is the **LC UPC Patch Cable, OM4**.

	LC UPC Patch Cable, OM4	SC APC Patch Cable, OS2
Fiber count	2 fibers	1 fibers
Wavelength	850/1300nm	1310/1550nm
Insertion Loss	$\leq 0.3\text{dB}$	$\leq 0.3\text{dB}$
Return Loss	$\geq 30\text{dB}$	$\geq 60\text{dB}$

The **SC APC Patch Cable, OS2** is more suitable choice for the faculty due to its compatibility with long-distance single-mode fiber applications, making it ideal for connecting buildings or linking to an ISP. It supports high-speed data transmission over extended distances with its wavelength range to ensuring reliable connectivity for the faculty's infrastructure. Moreover, its high return loss indicates excellent signal quality. Overall, the SC APC Patch Cable aligns better with the faculty's need for scalable and robust networking.

4.3.4 Question and Answer in Choosing Networking Devices

1. Are you surprised by the price? How were you surprised?

The total cost of RM1,024,812 for the network setup is not surprising because we have chosen the best devices to ensure performance, reliability and scalability. For example, premium equipment like Cisco Catalyst 9130AX access points and Dell PowerEdge R750 servers were selected to meet the demands of high-speed connectivity, large user capacity and advanced functionality. High-quality devices are essential for supporting a stable and efficient network infrastructure, especially in an educational environment where labs, classrooms and offices rely on uninterrupted access.

Additionally, the budget was calculated to balance quality and cost-effectiveness. Every component, from routers and switches to patch panels and cables, was evaluated for its ability to meet the requirements without unnecessary overspending. By investing in reliable and durable devices while maintaining financial discipline, the setup ensures both immediate performance and long-term sustainability, making the overall cost reasonable and well-justified.

2. Have you ever considered cost as a factor for choosing networking devices?

When choosing networking devices, cost was not our main focus but we will always survey for the most affordable and quality devices. Our priority was to identify devices that best met our project needs and enhanced network efficiency within our budget. We thoroughly assess different options and select those that provide the ideal balance of performance, affordability, and alignment with our project objectives.

However, the cost is still an important consideration in selecting networking devices, even though it wasn't the primary focus. Here's a detailed explanation why we conclude that the cost is partially one of the factors for choosing networking devices:

1. Balancing Cost with Functionality

We have conducted surveys and comparisons across brands to identify devices that provide us the best balance of affordability and performance. For example, the router, Ubiquiti EdgeRouter 12 was selected for its advanced features like 12 Ethernet ports and higher

throughput while being cost-effective for large-scale setups compared to alternatives like the Cisco Catalyst, which is significantly more expensive. For the patch panels, the BELNET CAT6 24-Port Patch Panel was chosen over the Panduit Mini-Com® due to its lower price (RM 378 vs. RM 627.13) while still meeting project requirements for cable management and functionality. We ensure that devices fulfilled project objectives like supporting high-speed networks or handling high user capacity without exceeding the budget.

2. Strategic Cost Management

We make sure the devices we choose stay within the budget. For instance, the Cat6 Ethernet Cable, Bulk Mediabridge 1000-foot rolls were highlighted as more economical for extensive wiring across labs. For the Smartboards, The Huawei IdeaHub S2 was selected over Canon's option due to its lower price (RM 15,000 vs. RM 20,000-25,000), even though both provided interactive teaching capabilities. While cost wasn't the only consideration, it helped narrow choices to devices that offered value while avoiding unnecessary premium features.

3. Understanding the Project's Scope

With the requirement to equip multiple labs and offices, the cost became crucial in scaling the network effectively. Devices with scalability and high utility per cost unit were favoured, such as the Cisco Catalyst 9130AX series Access Points. These were selected despite being slightly more expensive than Aruba options because they supported a larger number of clients (over 1,000) and offered additional features like AI-enhanced analytics.

4. Long-Term Cost Implications

Beyond the initial purchase price, your team considered devices that minimized long-term costs. The Dell PowerEdge R750 server was chosen despite its higher upfront cost because of its superior performance, scalability, and management features, which reduce maintenance costs over time. While for the Fiber Optic Cables, the SC APC Patch Cable was selected for its high return loss ($\geq 60\text{dB}$), ensuring minimal signal loss and reliable long-distance data transmission, reducing the need for frequent replacements or troubleshooting.

While cost wasn't your primary focus, we acknowledge that it played a key role in ensuring the project stayed within budget. We compared multiple options and strategically opted for devices that achieved the best balance of quality, functionality, and affordability.

4.3.5 Reflection on Task 3

Task 3 provided valuable insights into the importance of selecting the right networking devices while balancing performance, functionality and cost-effectiveness. Initially, we were surprised by the high cost of networking equipment but through extensive research and comparisons, we realised that investing in high-quality and scalable devices is crucial for long-term efficiency. We learned to evaluate different brands and models, considering factors such as network capacity and future scalability, ensuring that our selections aligned with the faculty's needs.

Another key takeaway from this task was the process of cost management. While our priority was network performance and reliability, we had to strategically allocate the budget to avoid unnecessary overspending. By analysing various devices such as routers, switches, patch panels and cabling, we identified cost-effective options that met both technical and financial constraints. This experience helped us understand the trade-offs between affordability and quality in network planning.

Overall, Task 3 has enhanced our ability to do research, compare and justify network hardware choices based on both technical specifications and budget considerations. This hands-on experience has strengthened our practical knowledge of network setup.

4.4 Task 4 LAN and WAN Connection

4.4.1 Identify Work Areas on Floor Plan

Work Area at Ground Floor



Figure 4.4.1.1 Work Area at Ground Floor

The Work Area 1 and 3 is General Lab which is 14m in length and 10m in width. This lab is to provide a space for students to work on general computing tasks, coursework and software-based projects. Each lab is equipped with a total of 31 PC workstations including the one at the lecturer's table. Consequently, the total of 24-port switches needed by the general labs is three switches to connect to all the workstations.

Next, for Work Area 2 that shown in Figure 4.4.1, is a Cisco Networking Lab which has a purpose to offer hands-on experience in network setup, configuration and troubleshooting,

preparing students for networking careers. This lab also consists of 30 workstations and a total of 5 server racks for students to learn about networking setup. In this lab, there is also a projector that can show teaching material easily.

At the bottom right of Figure 4.4.1, there is work area 4 which is Embedded Lab. This lab is built for enabling students to learn hardware-software integration and embedded system development, focusing on IoT and real-time applications. It contains 32 workstations including lecturer's PC and projector for easy teaching in the class.

Work Area of First Floor

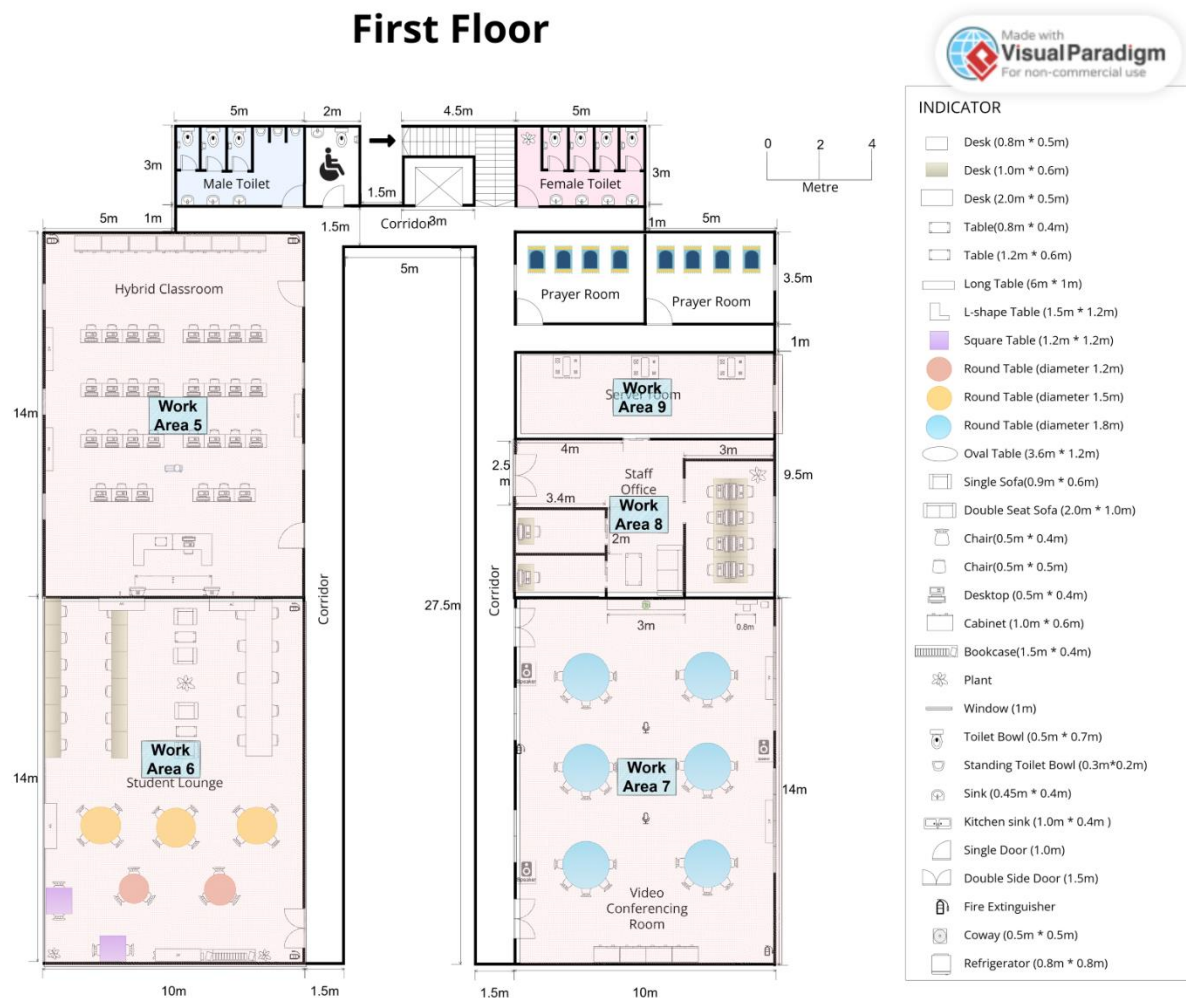


Figure 4.4.1.2 Work Area at First Floor

Work area 5 shown in Figure 4.4.2 is a hybrid classroom which is designed for interactive learning that combines physical and virtual sessions. It is equipped with 31 PC workstations and advanced technology such as projector, whiteboard and sound system, ensuring an engaging learning environment for both in-person and remote participants.

In Figure 4.4.2, the bottom left shows a student lounge space for students to relax and collaborate. It features comfortable seating options such as sofas, round tables for group discussions and a cozy environment to encourage informal interactions and socialization.

Meanwhile, work area 7 is a video conferencing room which serves as a dedicated space for virtual meetings, presentations and collaborative sessions, supporting remote connectivity with high-speed internet and audio-visual tools. It is equipped with round tables, ergonomic chairs and high-quality video conferencing systems.

Next, work area 8 shown in Figure 4.4.2 is a staff office which is designed for faculty and administrative tasks. It includes 8 PC workstations in common space while each small workspace has 1 PC workstation to ensure a professional workspace for staff, enhancing productivity and organization.

Lastly, work area 9 is a server room dedicated to housing and managing critical IT infrastructure. It includes 3 server racks and cooling systems to maintain optimal temperature and performance. This secure area is designed to ensure reliable connectivity and data management for the entire floor.

4.4.2 Distribution and Connection of Devices

Network Distribution Diagram

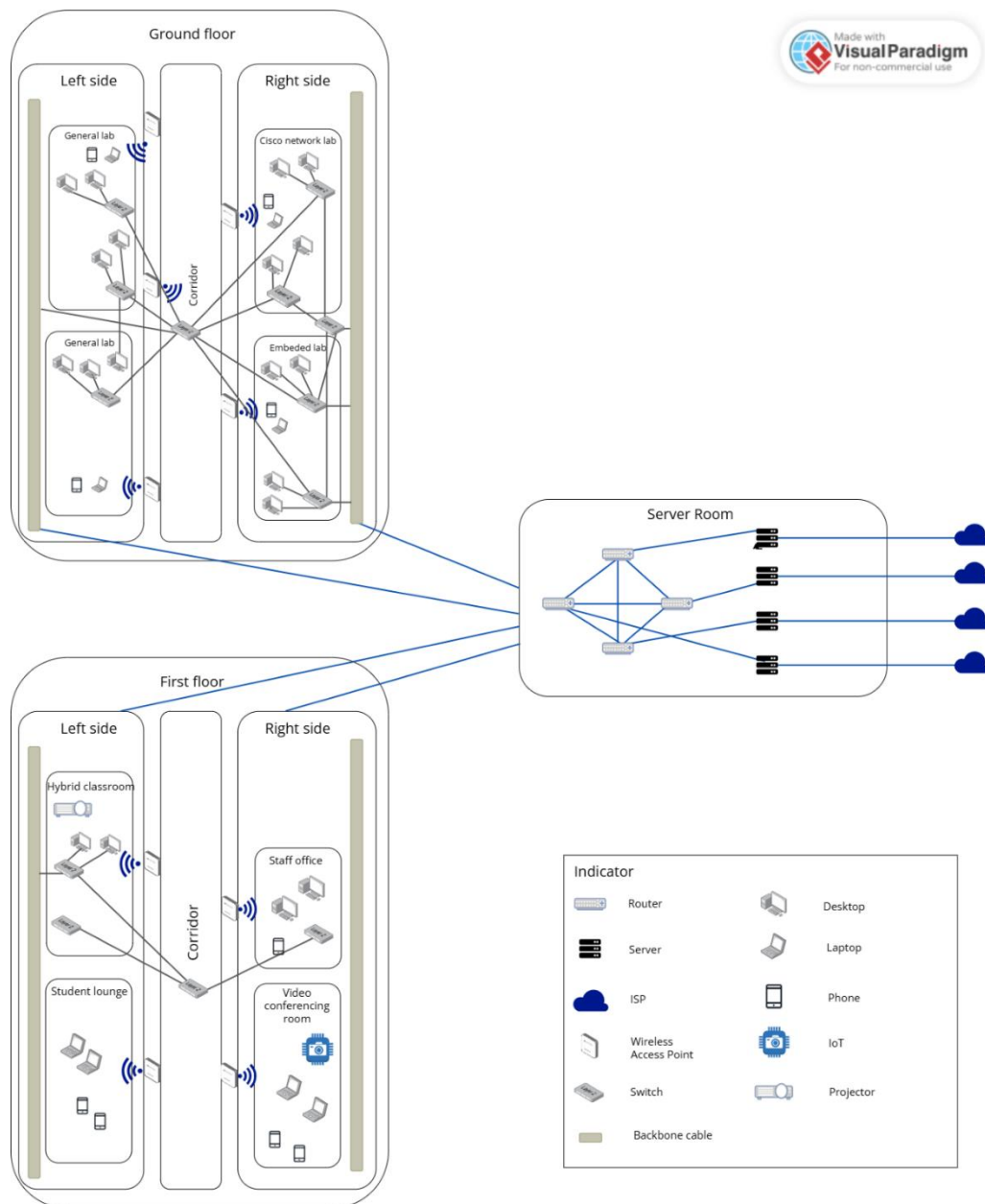


Figure 4.4.2.1 Network Distribution Diagram

In Figure 4.4.3, the diagram illustrates the network's layout and connectivity for both the ground and first floors of the building, ensuring seamless integration of wired and wireless devices. On each floor, switches are deployed to connect wired devices such as desktops and IoT devices in areas like general labs, Cisco Network Lab, Embedded Lab, hybrid classroom, staff office and video conferencing room. Wireless access points are strategically placed to provide Wi-Fi coverage, enabling mobile and wireless device connectivity. Each switch serves

as a local hub, consolidating connections and maintaining efficient data flow within its designated area.

All switches on the ground and first floors are linked to the server room via backbone cables. Which ensures high-speed and reliable data transfer between floors and core switch. In the server room, servers are connected to switches for managing computing tasks, data storage and application hosting. These switches are further linked to a router, which connects to the ISP for internet access. This hierarchical structure ensures a centralized yet flexible network arrangement, supporting both high-performance wired connections and robust wireless communication throughout the building.

For a clearer depiction of the network distribution on the ground and first floors, refer to figures 4.4.4 and 4.4.5. The diagram outlines how wireless access points, switches and devices in various sections of the building connect through backbone cables to the switches and router housed in the server room. The connections are simplified to highlight the overall structure, while Figure 4.4.6 provides a detailed illustration of the server room.

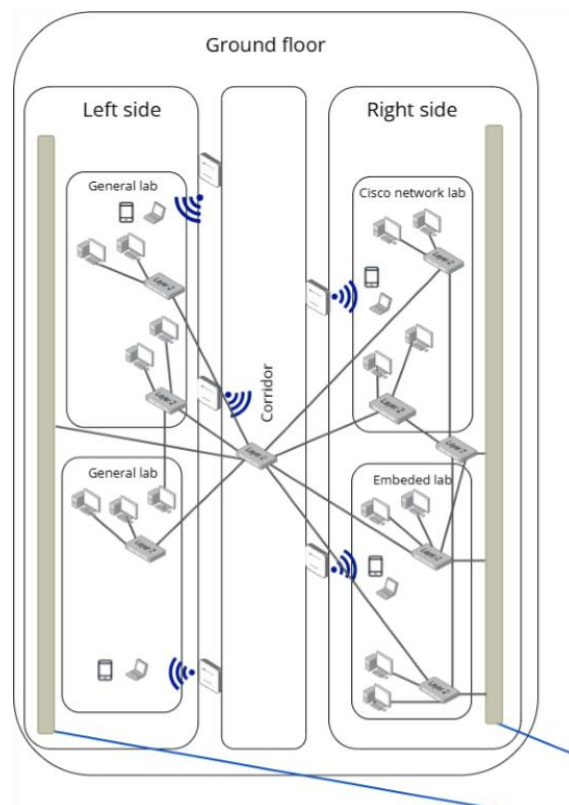


Figure 4.4.2.3 Network Distribution Diagram – Ground Floor

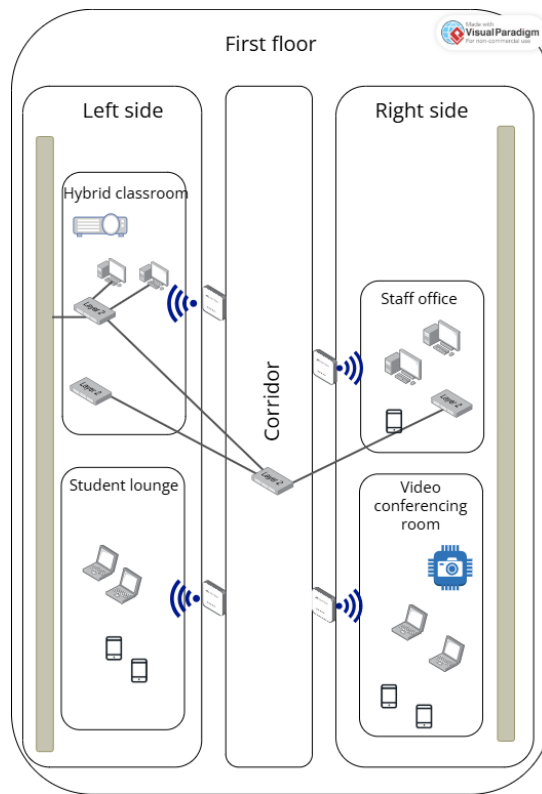


Figure 4.4.2.4 Network Distribution Diagram – First Floor

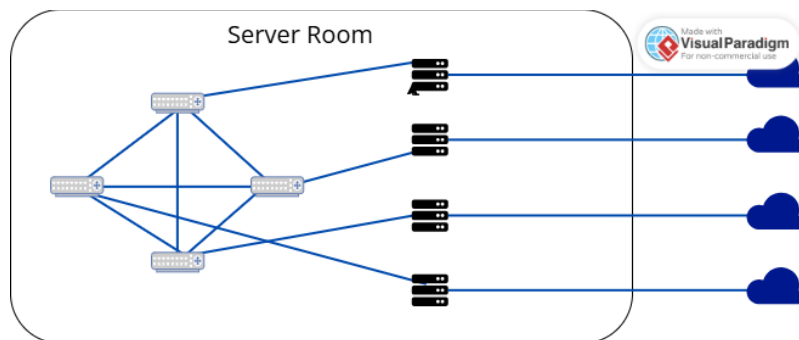


Figure 4.4.2.5 Network Distribution Diagram – Server Room

The depicted diagram illustrates the network distribution across the ground floor, first floor and server room with an emphasis on the router and server configuration. In this design, four routers are integrated into the network within the server room, each independently connected using fiber optic cables. This configuration is intended to prevent a complete network outage in the event of a single router failure.

To enhance network resilience, the routers are arranged in a mesh topology which allows seamless functionality even if one router encounters issues. Across the building, a single switch is deployed in most rooms to establish localized subnets, ensuring stability within each subnet. Moreover, one switch is redundantly connected to two routers which provide a failover mechanism to maintain network operations during potential disruptions or emergencies. This strategic setup ensures robust connectivity and fault tolerance across all network layers.

Floor Diagram

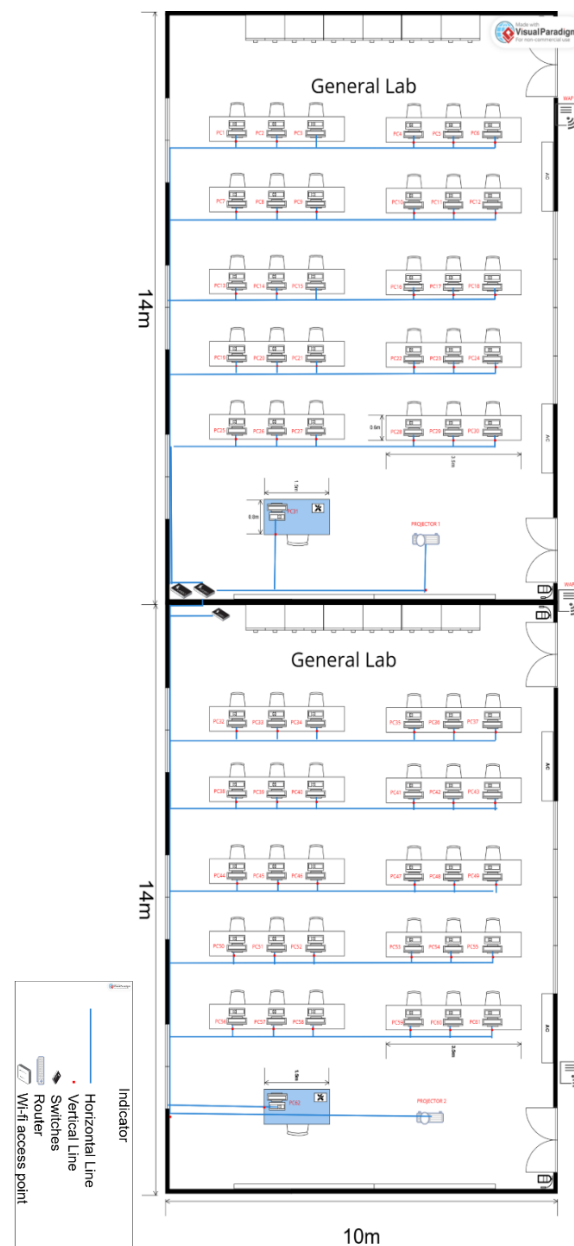


Figure 4.4.2.6 Cable Distribution Diagram – General Labs

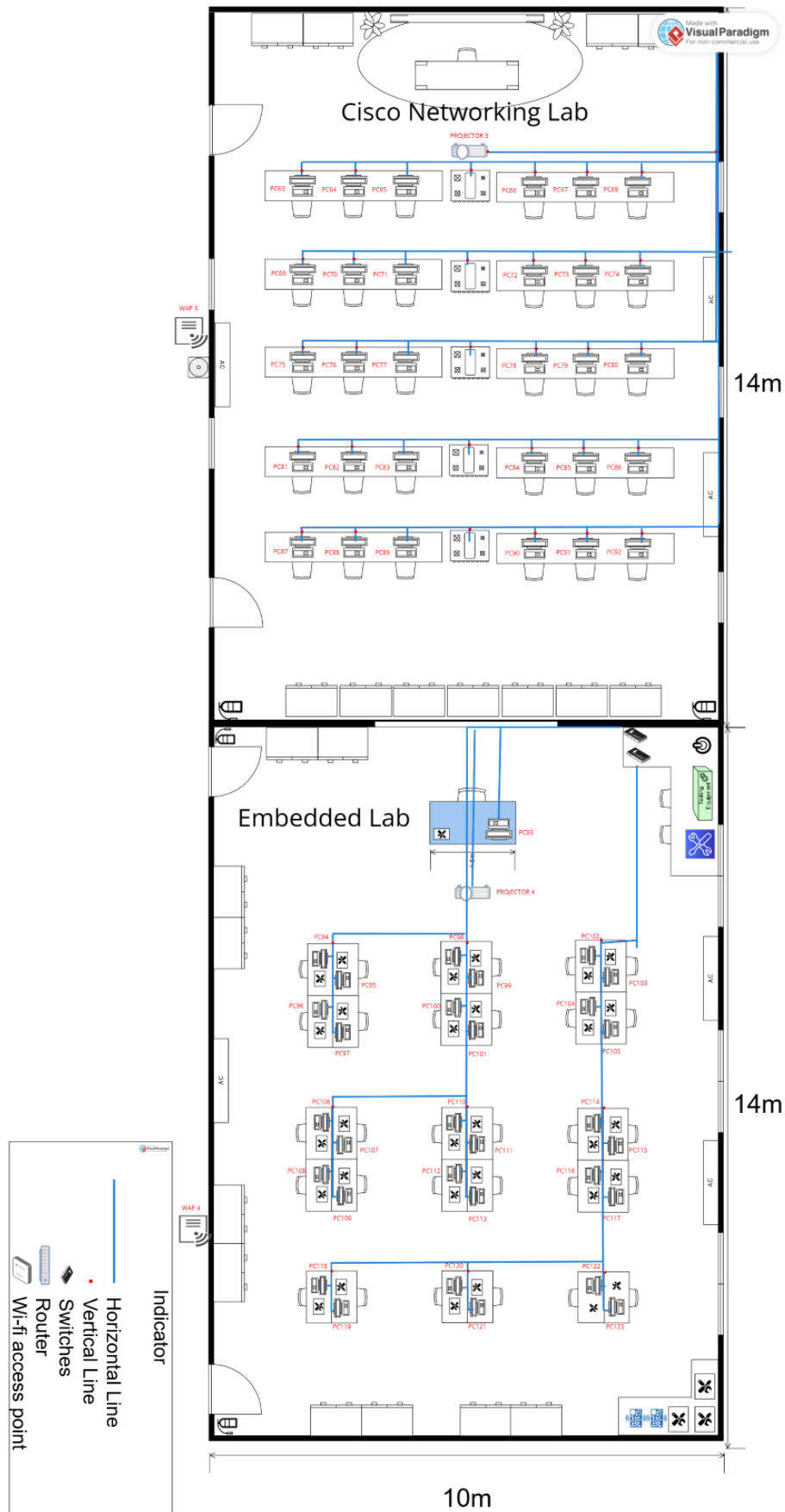


Figure 4.4.2.7 Cable Distribution Diagram – Cisco Networking Lab and Embedded Lab

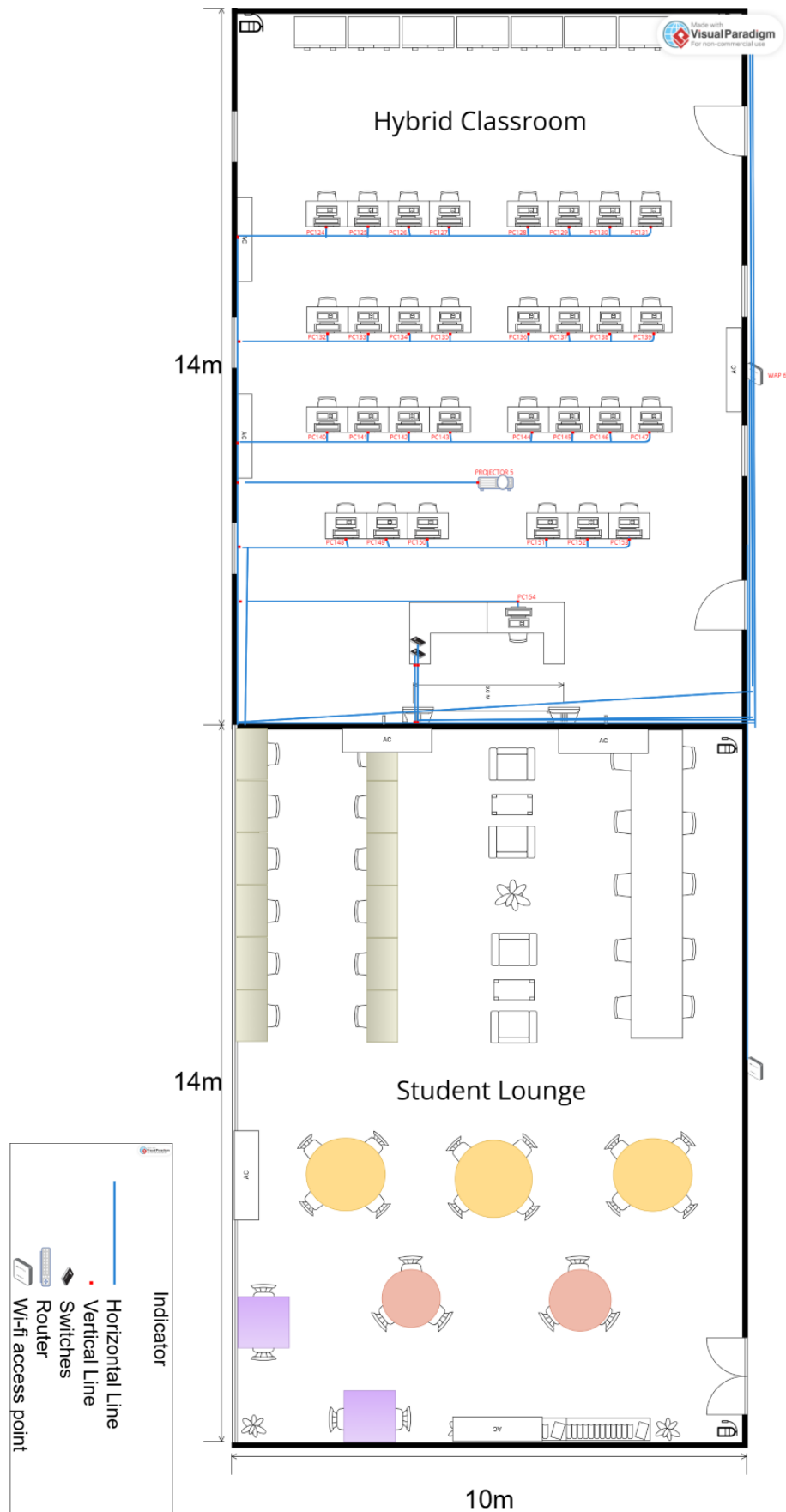


Figure 4.4.2.8 Cable Distribution Diagram – Hybrid Classroom and Student Lounge

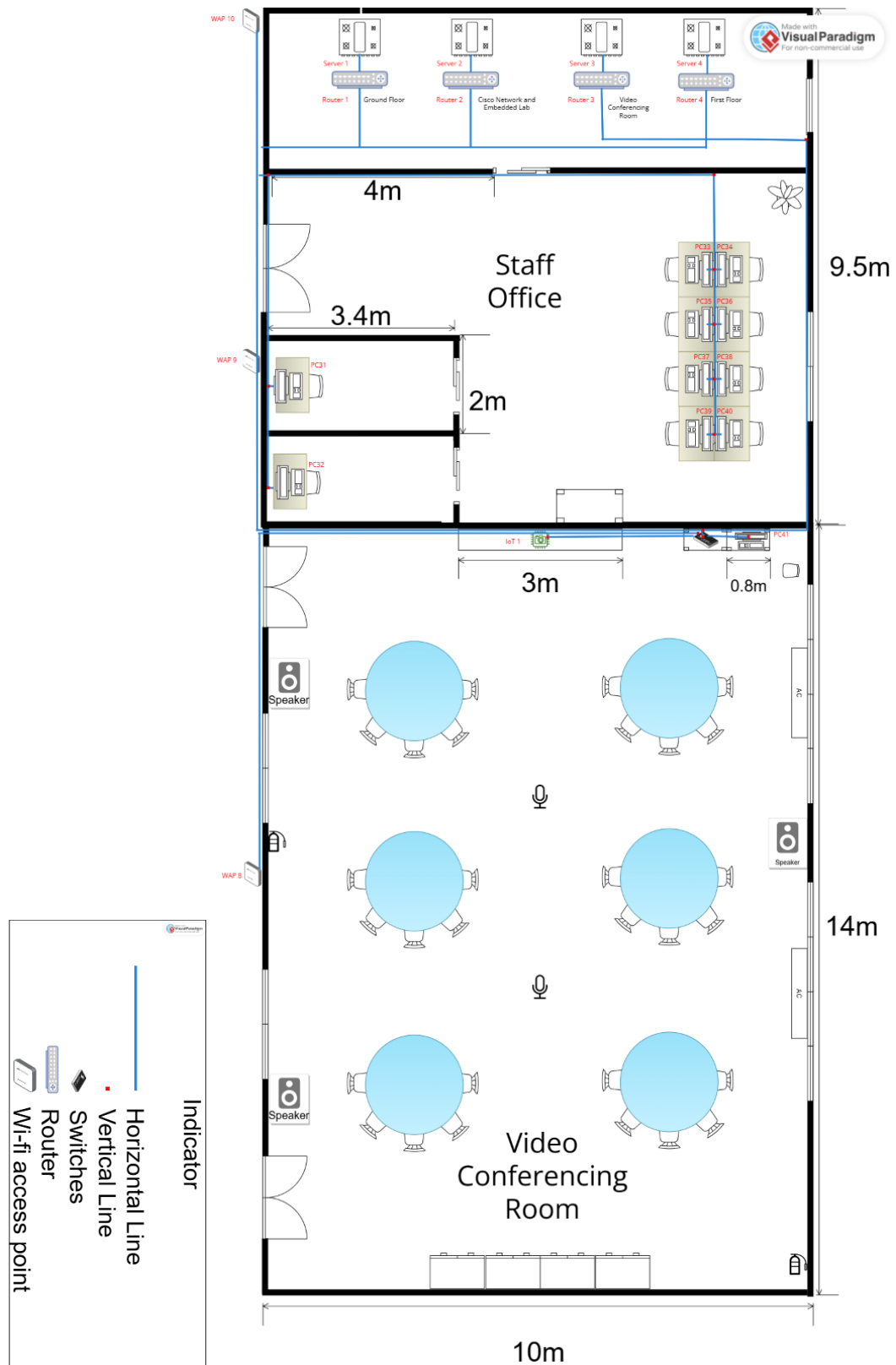


Figure 4.4.2.9 Cable Distribution Diagram –Staff Office, Server Room and Video Conferencing Room

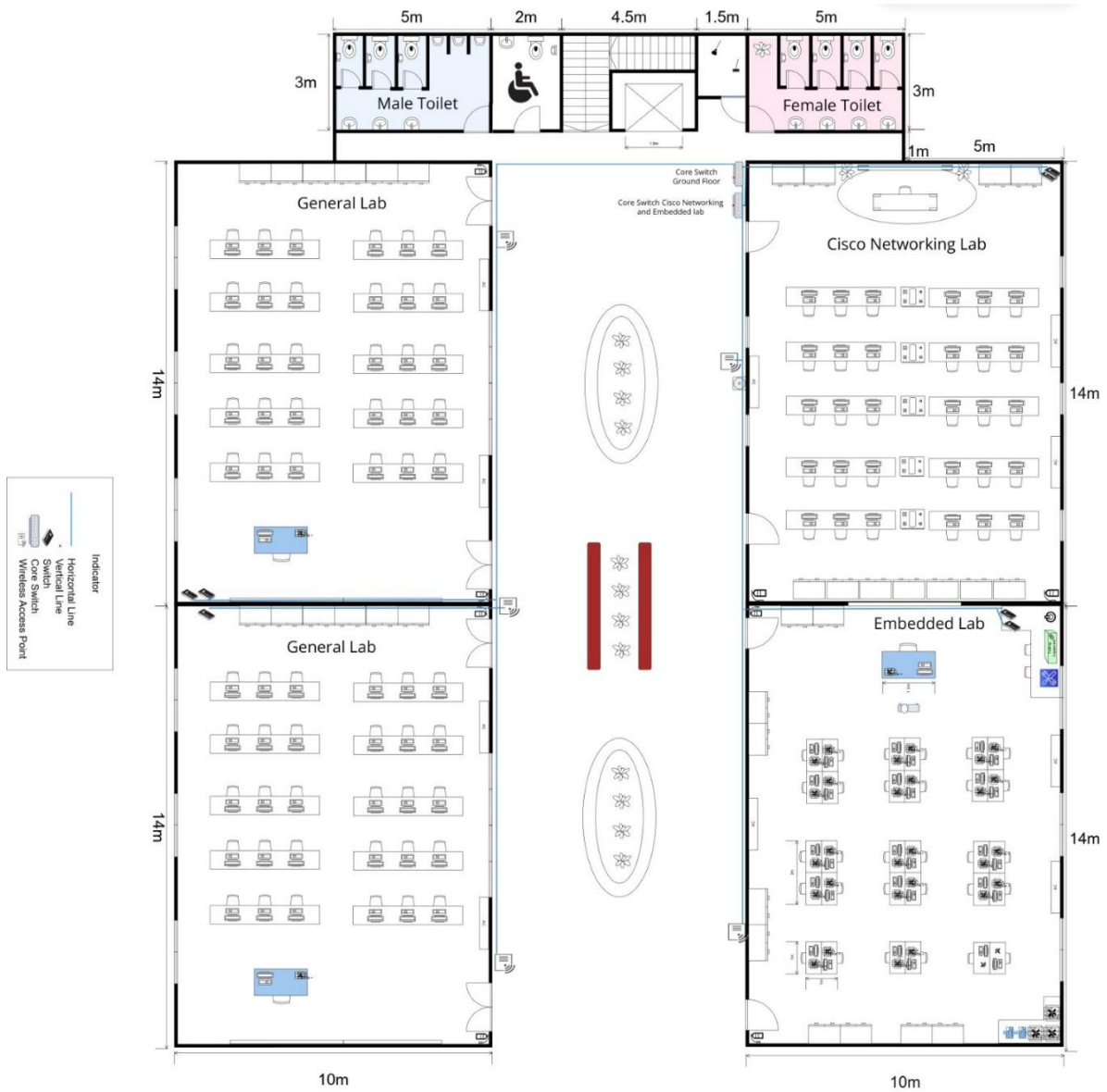


Figure 4.4.2.10 Cable Distribution Diagram – Ground Floor

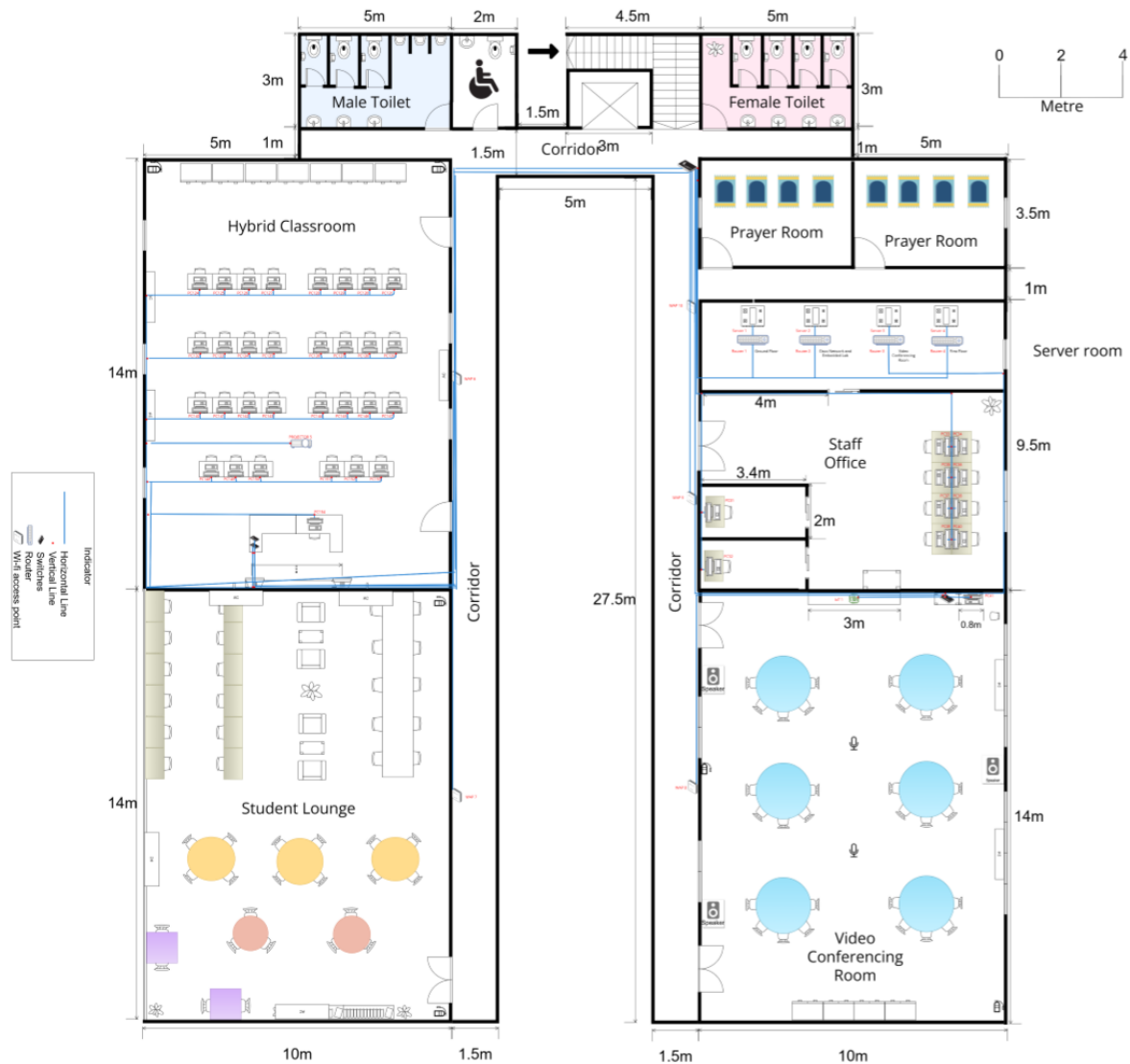


Figure 4.4.2.11 Cable Distribution Diagram – First Floor

The floor diagrams illustrate the network connections within the faculty building, showing how switches on each floor are interconnected and linked via backbone connections. Each floor has a core switch that consolidates connections from all individual switches on that floor. The core switches are then linked to dedicated routers, establishing a robust and efficient network. Additionally, key areas such as the Embedded Lab, Cisco Networking Lab, and the Video Conferencing Room are highlighted in the diagrams, showing their direct or indirect connections to the routers.

4.4.3 Number of Connections, Patch Cords and Switch Ports Needed

Area	Number of Switch Ports Needed
General Lab 1	36
General Lab 2	34
Cisco Network Lab	39
Embedded Lab	35
Hybrid Classroom	36
Video Conference Room	5
Staff Office	10
Server Room	5
Total	200

Table 4.4.1: Number of Switch Ports Needed

Area	Number of Switch Ports Needed
General Lab 1	36
General Lab 2	34
Cisco Network Lab	39
Embedded Lab	35
Hybrid Classroom	36
Video Conference Room	5
Staff Office	10
Server Room	5
Total	200

Table 4.4.2: Number of Patch Cords Needed

Area	Number of Switch Ports Needed
General Lab 1	72
General Lab 2	68
Cisco Network Lab	78
Embedded Lab	70
Hybrid Classroom	72
Video Conference Room	10
Staff Office	20
Server Room	10
Total	400

Table 4.4.3: Number of RJ45 Connector Needed

4.4.4 Identify Cable Types and Length

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
General Lab 1	184.5	27.3	211.8	1129.6
General Lab 2	191.5	26.5	218	1162.67
Cisco Network Lab	191	29.5	220.5	1176
Embedded Lab	74.1	16.5	90.6	483.2
Ground Floor Room Total			740.9	3951.47
First Floor				
Hybrid Classroom	82.8	47.05	129.85	692.53
Video Conference Room	21.80	5.00	26.80	142.94
Server Room	17.40	5.60	23.00	122.67
Staff Office	27.2	13.1	40.3	214.93
First Floor Room Total			213.75	1173.07

Path	Length(m)	Price of cables used (RM)
Ground Floor		
Server room to Core Switch Ground Floor	15	80
Server room to Core Switch Cisco Networking Lab and Embedded Lab	15	80

Wifi Access Point to switches	59	314.7
Switches to Core Switch	117	624
Ground Floor Room Total	206	1098.7
Total - Ground Floor		
Total length of wire to connect the rooms (m)	206	1098.7
Total length of wire to connect end devices to switches in rooms (m)	740.9	3951.47
Total length of wire needed (m)	946.9	5050.17
First Floor		
Wi-Fi Access Point to Switches	37.4	59.84
Hybrid Classroom to core switch	46.4	74.24
Video Conferencing Room to core switch	14	22.4
Staff Office to Core Switch	18	28.8
Server Room to Core Switch	7.2	11.52
First Floor Total	123	196.80
Total First Floor		
Total length of wire to connect the room (m)	123	196.80
Total length of wire to connect end devices to switches in rooms (m)	213.75	1173.07
Total length of wire needed (m)	336.75	1369.87
Total (Ground Floor + First Floor) (m)	1283.65	6420.04

Choosing the right cables is important for good connectivity. After discussing, we'll use both CAT6 twisted cables and fiber optic cables in the building.

We'll use CAT6 twisted cables for end devices connections in labs and classroom such as desktops, projector and other network end user devices. These cables can handle high-speed data up to 10 gigabits per second since they have a high bandwidth. CAT6 cables have thicker copper wires that help with heat dissipation, making them good for Power over Ethernet (PoE), which reduces cabling time and costs. The specific type of CAT6 chosen is Unshielded Twisted Pair (UTP), known for reducing electromagnetic interference caused by natural or human-made sources.

While fiber optic cables are designated for interconnecting routers within the server room and extending the connection to the Internet Services Provider (ISP). By utilizing light for data transmission, fiber optic cables facilitate high-speed data transfer while minimizing signal degradation. With a bandwidth capacity up to 10Gbps, these cables are well-suited for handling significant data volumes. Moreover, fiber optic cables are highly effective at transmitting signals over longer distances, which guarantees fast and reliable data transmission.

In summary, our building requires a specific cable length, including both CAT6 and fiberoptic cables. This will ensure seamless internal connectivity among devices within the building and a reliable connection to the Internet Service Provider (ISP).

4.4.5 Reflection Task 4

This task helped us gain a deeper understanding of network design and implementation. We learned how to structure LAN and WAN networks, ensuring efficient data flow and reliable connectivity. Understanding how to connect all the network devices allowed us to design a more stable and scalable network system. We explored cable management which included selecting the right cable types, estimating lengths, and planning routing paths for better performance and cost efficiency. We also gained insight into how switch ports, patch cords, and backbone connections play a crucial role in maintaining a strong and efficient network.

In addition, this task helped us develop skills in network planning and resource estimation. We learned how to calculate the required number of connections, cable lengths, and networking equipment based on real-world needs. Understanding the difference between CAT6 cables for local connections and fibre optic cables for high-speed data transfer allowed us to see how different technologies are applied depending on network requirements.

Overall, this task improved our knowledge of network architecture, hardware selection, and efficient resource management. It provided us with valuable practical experience in designing a well-structured, cost-effective, and reliable network infrastructure while considering both technical requirements and real-world constraints.

4.5 Task 5

4.5.1 Network Address Given

Group	Network Address
12.3	192.164.0.0/8

The given network address of our group is 192.164.0.0/8. It provides over 16 million IPs (2^{24}) which making it suitable for large-scale networks with thousands of devices. This ensures future scalability for massive expansion.

Network Address details:

IP address (decimal)	192.164.0.0
IP address (binary)	11000000 10100100 00000000 00000000
Subnet mask (decimal)	255.0.0.0
Subnet Mask (binary)	11111111 00000000 00000000 00000000

We divided the subnet mask into the network portion (8 bits) and host portion (24 bits). By using the AND operation of the IP address with the subnet mask, we identified the following:

Network Address	192.0.0.0
Broadcast Address	192.255.255.255
Range of Usable Addresses	192.0.0.1 – 192.255.255.254

4.5.2 Divide it in the best possible way for network

By using the Classless Inter-Domain Routing (CIDR) method, we decided to divide the network into separate subnets for each work area outlined in task 4, except for the server room and staff office which share the same subnet. A total of 8 subnets are needed to accommodate the design of our network. Thus, we need to borrow 3 bits from the host portion of the IP address, as

2³2³= 8 subnets are necessary to meet the network's segmentation requirements.

By borrowing these 3 bits, the subnet mask changed from /8 to /11.

Network Portion = Original 8 bits + 3 borrowed subnet bits = 11 bits

Host Portion = Total 32 bits – 11 Network Portion bits = 21 bits

This approach ensures that each subnet and devices and device have sufficient IP addresses while maintaining future scalability and efficient resource allocation across the faculty's network infrastructure.

Divide subnet to 8 subnets:

Subnet Network	Network portion	Subnet bits	Host portion		IP address
0	1100 0000	000	0 0000 0000 0000 0000 0000	Network Address	192.0.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.31.255.255
1	1100 0000	001	0 0000 0000 0000 0000 0000	Network Address	192.32.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.63.255.255
2	1100 0000	010	0 0000 0000 0000 0000 0000	Network Address	192.64.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.95.255.255
3	1100 0000	011	0 0000 0000 0000	Network Address	192.96.0.0

			0000 0000		
			1 1111 1111 1111 1111 1111	Broadcast Address	192.127.255.255
4	1100 0000	100	0 0000 0000 0000 0000 0000	Network Address	192.128.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.159.255.255
5	1100 0000	101	0 0000 0000 0000 0000 0000	Network Address	192.160.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.191.255.255
6	1100 0000	110	0 0000 0000 0000 0000 0000	Network Address	192.192.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.223.255.255
7	1100 0000	111	0 0000 0000 0000 0000 0000	Network Address	192.224.0.0
			1 1111 1111 1111 1111 1111	Broadcast Address	192.255.255.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Custom subnet mask
0	Server Room and Staff Office	192.0.0.0	192.31.255.255	192.0.0.1 - 192.31.255.254	/11 255.224.0.0
Ground Floor					
1	General Lab 1	192.32.0.0	192.63.255.255	192.32.0.1 - 192.63.255.254	/11 255.224.0.0
2	General Lab 2	192.64.0.0	192.95.255.255	192.64.0.1 - 192.95.255.254	/11 255.224.0.0
3	Cisco Networking Lab	192.96.0.0	192.127.255.255	192.96.0.1 - 192.127.255.254	/11 255.224.0.0
4	Embedded Lab	192.128.0.0	192.159.255.255	192.128.0.1 - 192.159.255.254	/11 255.224.0.0
First Floor					
5	Hybrid Classroom	192.160.0.0	192.191.255.255	192.160.0.1 - 192.191.255.254	/11 255.224.0.0
6	Student Lounge	192.192.0.0	192.223.255.255	192.192.0.1 - 192.223.255.254	/11 255.224.0.0

7	Video Conferencing Room	192.224.0.0	192.255.255.255	192.224.0.1 - 192.255.255.254	/11 255.224.0.0
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Network distribution for workstation in different work areas.

Ground Floor:

1. General Lab 1

Device	IP Address
PC1	192.32.0.1
PC2	192.32.0.2
PC3	192.32.0.3
PC4	192.32.0.4
PC5	192.32.0.5
PC6	192.32.0.6
PC7	192.32.0.7
PC8	192.32.0.8
PC9	192.32.0.9
PC10	192.32.0.10
PC11	192.32.0.11
PC12	192.32.0.12
PC13	192.32.0.13
PC14	192.32.0.14
PC15	192.32.0.15
PC16	192.32.0.16
PC17	192.32.0.17
PC18	192.32.0.18
PC19	192.32.0.19
PC20	192.32.0.20
PC21	192.32.0.21
PC22	192.32.0.22

PC23	192.32.0.23
PC24	192.32.0.24
PC25	192.32.0.25
PC26	192.32.0.26
PC27	192.32.0.27
PC28	192.32.0.28
PC29	192.32.0.29
PC30	192.32.0.30
PC31	192.32.0.31
Projector 1	192.32.0.32
WAP 1	192.32.0.33
WAP 2	192.32.0.34

2. General Lab 2

Device	IP Address
PC32	192.64.0.1
PC33	192.64.0.2
PC34	192.64.0.3
PC35	192.64.0.4
PC36	192.64.0.5
PC37	192.64.0.6
PC38	192.64.0.7
PC36	192.64.0.8
PC40	192.64.0.9
PC41	192.64.0.10
PC42	192.64.0.11
PC43	192.64.0.12
PC44	192.64.0.13
PC45	192.64.0.14
PC46	192.64.0.15

PC47	192.64.0.16
PC48	192.64.0.17
PC49	192.64.0.18
PC50	192.64.0.19
PC51	192.64.0.20
PC52	192.64.0.21
PC53	192.64.0.22
PC54	192.64.0.23
PC55	192.64.0.24
PC56	192.64.0.25
PC57	192.64.0.26
PC58	192.64.0.27
PC59	192.64.0.28
PC60	192.64.0.29
PC61	192.64.0.30
PC62	192.64.0.31
Projector 2	192.64.0.32
WAP 3	192.64.0.33

3. Cisco Networking Lab

Device	IP Address
PC63	192.96.0.1
PC64	192.96.0.2
PC65	192.96.0.3
PC66	192.96.0.4
PC67	192.96.0.5
PC68	192.96.0.6
PC69	192.96.0.7
PC70	192.96.0.8
PC71	192.96.0.9
PC72	192.96.0.10

PC73	192.96.0.11
PC74	192.96.0.12
PC75	192.96.0.13
PC76	192.96.0.14
PC77	192.96.0.15
PC78	192.96.0.16
PC79	192.96.0.17
PC80	192.96.0.18
PC81	192.96.0.19
PC82	192.96.0.20
PC83	192.96.0.21
PC84	192.96.0.22
PC85	192.96.0.23
PC86	192.96.0.24
PC87	192.96.0.25
PC88	192.96.0.26
PC89	192.96.0.27
PC90	192.96.0.28
PC91	192.96.0.29
PC92	192.96.0.30
Projector 3	192.96.0.31
WAP 4	192.96.0.32

4. Embedded Lab

Device	IP Address
PC93	192.128.0.1
PC94	192.128.0.2
PC95	192.128.0.3
PC96	192.128.0.4
PC97	192.128.0.5
PC98	192.128.0.6

PC99	192.128.0.7
PC100	192.128.0.8
PC101	192.128.0.9
PC102	192.128.0.10
PC103	192.128.0.11
PC104	192.128.0.12
PC105	192.128.0.13
PC106	192.128.0.14
PC107	192.128.0.15
PC108	192.128.0.16
PC109	192.128.0.17
PC110	192.128.0.18
PC111	192.128.0.19
PC112	192.128.0.20
PC113	192.128.0.21
PC114	192.128.0.22
PC115	192.128.0.23
PC116	192.128.0.24
PC117	192.128.0.25
PC118	192.128.0.26
PC119	192.128.0.27
PC120	192.128.0.28
PC121	192.128.0.29
PC122	192.128.0.30
PC123	192.128.0.31
Projector 4	192.128.0.32
WAP 5	192.128.0.33

Ground floor with label:

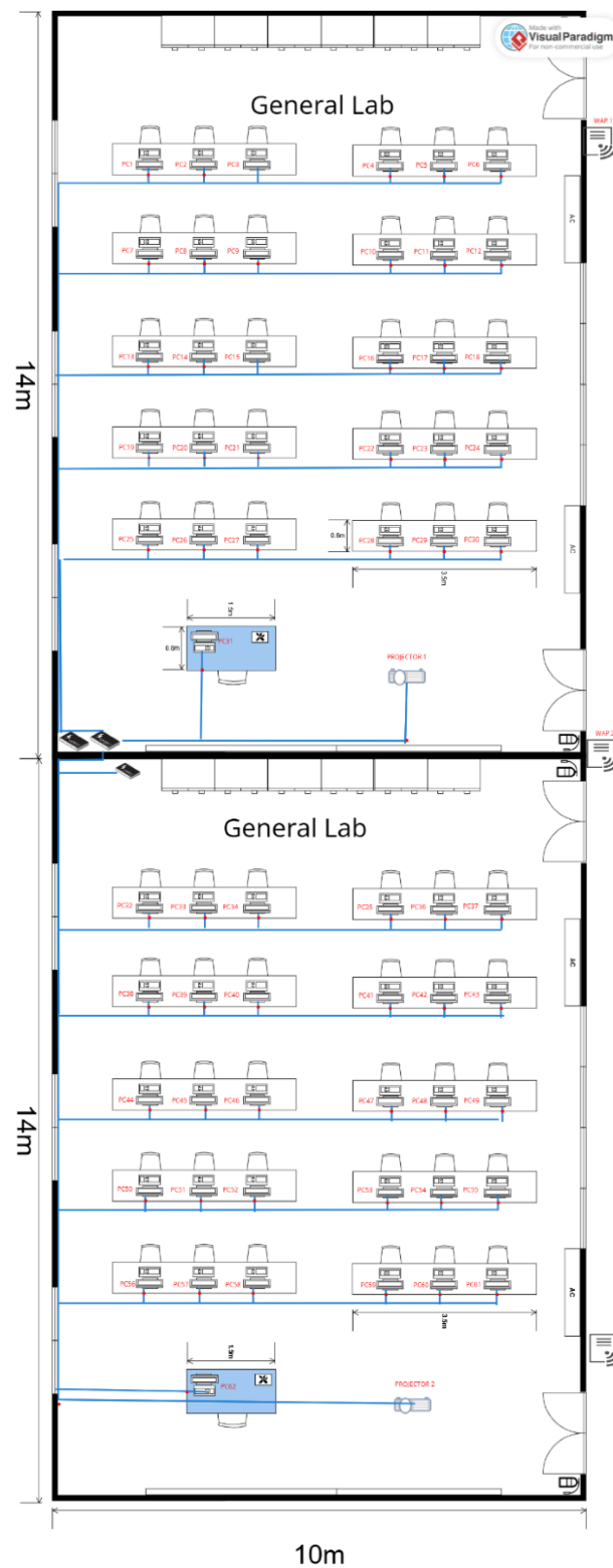


Figure 4.5.2.1 General Lab 1 & 2 with label

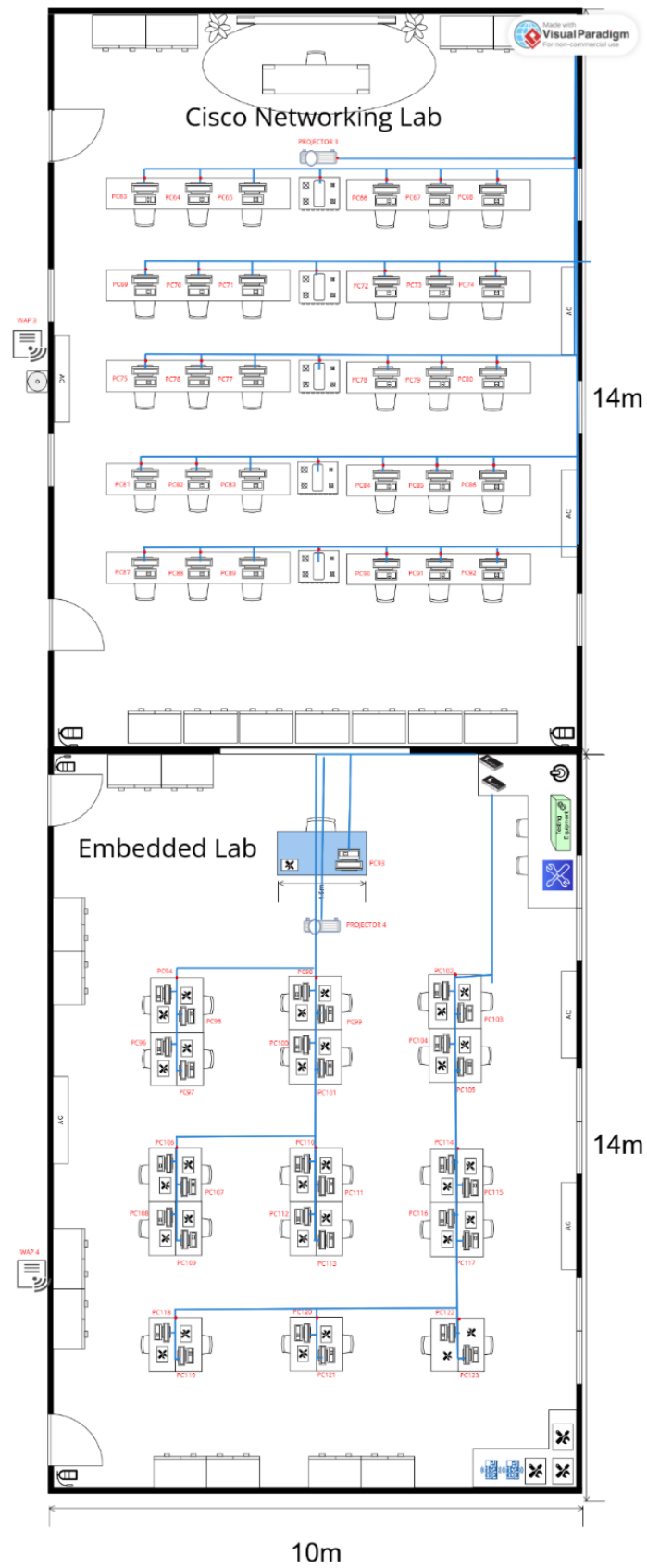


Figure 4.5.2.2 Cisco Networking Lab and Embedded Lab with label

First Floor:

5. Hybrid Classroom

Device	IP Address
PC124	192.160.0.1
PC125	192.160.0.2
PC126	192.160.0.3
PC127	192.160.0.4
PC128	192.160.0.5
PC129	192.160.0.6
PC130	192.160.0.7
PC131	192.160.0.8
PC132	192.160.0.9
PC133	192.160.0.10
PC134	192.160.0.11
PC135	192.160.0.12
PC136	192.160.0.13
PC137	192.160.0.14
PC138	192.160.0.15
PC139	192.160.0.16
PC140	192.160.0.17
PC141	192.160.0.18
PC142	192.160.0.19
PC143	192.160.0.20
PC144	192.160.0.21
PC145	192.160.0.22
PC146	192.160.0.23
PC147	192.160.0.24
PC148	192.160.0.25
PC149	192.160.0.26

PC150	192.160.0.27
PC151	192.160.0.28
PC152	192.160.0.29
PC153	192.160.0.30
PC154	192.160.0.31
Projector 5	192.160.0.32
WAP 6	192.160.0.33

6. Student Lounge

Device	IP Address
WAP 7	192.192.0.1

7. Video Conferencing Room

Device	IP Address
IoT 1	192.224.0.1
WAP 8	192.224.0.2

8. Staff Office

Device	IP Address
PC155	192.0.0.1
PC156	192.0.0.2
PC157	192.0.0.3
PC158	192.0.0.4
PC159	192.0.0.5
PC160	192.0.0.6
PC161	192.0.0.7
PC162	192.0.0.8
PC163	192.0.0.9
PC164	192.0.0.10
WAP 9	192.0.0.11

9. Server Room

Device	IP Address
Router 1	192.0.0.12
Router 2	192.0.0.13
Router 3	192.0.0.14
Router 4	192.0.0.15
Server 1	192.0.0.16
Server 2	192.0.0.17
Server 3	192.0.0.18
Server 4	192.0.0.19
WAP 10	192.0.0.20

First floor with label:

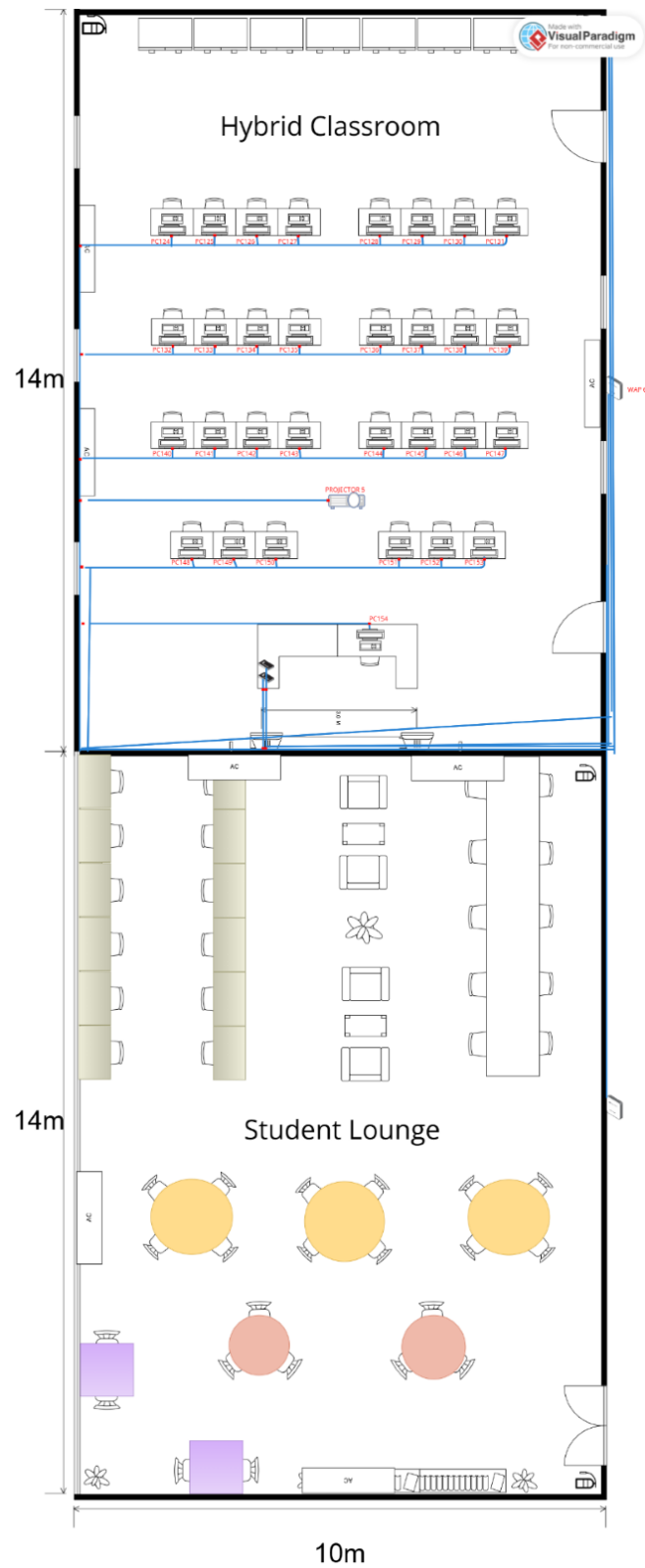


Figure 4.5.2.3 Hybrid Classroom and Student Lounge with label

4.5.1 Task 5 Reflection

Completing Task 5 was a collaborative effort that showcased the strengths and teamwork of our group. Each member contributed unique skills, whether it was in calculating subnetting, designing network layouts or managing documentation. We divided the responsibilities effectively. We separated tasks into 2 parts which is calculate subnetting and divide all IP address into subnets and distribute it into each device. This division allowed us to work efficiently and meet the deadlines while learning from each other's expertise.

One of the major lessons we learned during this task was the importance of proper subnetting in network design. By understanding how to allocate IP addresses efficiently, we ensured that different departments and work areas within the network had sufficient resources. We also realized how critical it is to plan for future scalability and network security while designing the IP structure. This task deepened our understanding of IP addressing, subnet masks and the practical application of these concepts in a real-world scenario.

Additionally, this task emphasized the value of communication and collaboration within a team. Regular discussions and meetings help us align our ideas and troubleshoot any issues that arose. We learned how to delegate tasks based on individual strengths and how to integrate everyone's work into a cohesive final submission. The process improved our problem-solving skills and taught us to approach complex technical tasks methodically.

Overall, Task 5 was an invaluable learning experience that not only strengthened our technical knowledge but also enhances our teamwork abilities. We now have a better grasp of network design principles, and the practical steps involved in creating efficient and secure networks. The task also highlighted the importance of thorough planning and collaboration, lessons that will benefit us in future projects and professional settings.

5.0 Conclusion

In conclusion, this project has provided our team with a comprehensive understanding of network communication, from theoretical concepts to practical implementation. Through the various tasks, we were able to apply our knowledge in designing an efficient, scalable and secure network infrastructure for the new faculty building. The hands-on experience with network devices, subnetting and configuration tools has enhanced our technical skills, preparing us for real-world networking challenges.

The collaborative nature of this project allowed us to develop strong teamwork and communication skills. By effectively distributing tasks and leveraging each member's strengths, we ensured that every aspect of the project was handled proficiently. Regular meetings and open discussions facilitated problem-solving and idea-sharing, contributing to the overall success of the project.

Furthermore, the project highlighted the importance of balancing technical requirements with budget constraints. We learned to make informed decisions when selecting networking devices, considering both performance and cost-effectiveness. This approach not only ensured the project's feasibility but also provided valuable insights into resource management and strategic planning.

Overall, this project has been a significant learning experience that has equipped us with essential networking knowledge and teamwork skills. The insights gained will be invaluable in our future academic and professional endeavours, helping us to approach complex projects with confidence and expertise.

6.0 Additional Considerations for Informed Decision-Making

Cost Optimization Through Bulk Purchasing and Vendor Negotiations

A highly effective way to optimize costs is through bulk purchasing and vendor negotiations, which can significantly reduce expenses while adding value. Many suppliers offer discounts for large orders, making it more cost-efficient to buy networking equipment in bundles rather than individually. By negotiating comprehensive package deal that includes routers, access points, switches, and cabling from a single supplier, organizations can achieve a 15% to 20% reduction in total costs. Additionally, some vendors provide free onsite support, extended warranties, and maintenance plans, which help minimize future repair and replacement expenses. These strategic negotiations not only lower initial investment costs but also contribute to long-term network stability and efficiency, ensuring a more sustainable and cost-effective infrastructure.

7.0 Team Member and Responsibilities

Team Member	Information and Responsibilities
 Name: Poh Lok Yee Matric Number: A23CS0262	I am the team leader that responsible for overseeing the project, managing task distribution, and ensuring smooth coordination among all members. My role involves leading discussions, making key decisions, and handling the design of the student lounge and first-floor layout. Additionally, I manage the network distribution for the ground floor and review important project components such as the feasibility assessment and selected questions.
 Name: Chau Ying Jia Matric Number: A23CS0213	I am responsible for designing the second floor and Embedded Lab. I actively contribute by drawing floorplan drafts and finalizing layouts. Additionally, I provide valuable input on software tools such as Packet Tracer, plans the network distribution for the first floor, and works on subnetting and network division tasks. I always help to check the project are well-documented and on track.
 Name: Sabrina Heng Wei Qi Matric Number: A23CS0265	I am the organized and detail-oriented member who is responsible for designing the Cisco Network lab and Hybrid Classroom. I play a crucial role in documenting discussions, structuring project content, and ensuring that our work is well-organized. I also lead meetings when needed, conducts research on required devices, and helps in planning and assessing the network structure.
 Name: Woo Cheng Shuan Matric Number: A23CS0283	I responsible for designing the General Lab and Video Conferencing Room while assisting in discussions on floor plan designs and selecting the necessary devices. My role also includes structuring the network plan and workstation placement, and technical accuracy. As a meeting chair in some sessions, I help facilitate discussions and decision-making processes.

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9.0 Appendices

9.1 Financial Budget

Device	Price per unit (RM)	Quantity	Total price (RM)
Router  Figure 9.1.1 Router	1,269	4	5,076
Switches  Figure 9.1.2 Switches	2,033	13	26,429
Patch Panels  Figure 9.1.3 Patch Panels	378	13	4,914
Servers  Figure 9.1.4 Server	30,895	4	123,580
Wi-Fi Access Point	3,300	10	33,000

 Figure 9.1.5 Wi-Fi Access Point			
Projector  Figure 9.1.6 Projector	1,799	5	8,995
Desktop  Figure 9.1.7 Desktop	4,652	164	762,928
SmartBoard  Figure 9.1.8 Smartboard	15,000	1	15,000
Uninterrupted Power Supply (UPS)  Figure 9.1.9	36,000	1	36,000

Uninterrupted Power Supply (UPS)			
Cat6 Ethernet Cable  Figure 9.1.10 Cat6 Ethernet Cable	80/15m	1500m	8,000
Connector  Figure 9.1.11 Connector	0.38	500	190
Fiber Optic Cables  Figure 9.1.12 Fiber Optic Cables	14/1m	50	700
		TOTAL	1,024,812
		Budget	5,000,000
		Remaining Budget	3,975,188
		Budget Usage	$(1,024,812/5,000,000)$ $\times 100\% = 20.50\%$

9.2 Meeting Minutes

MEETING MINUTES #1

DATE/TIME	4 Nov 2024 3pm		
LOCATION	Arked Angkasa		
AGENDA	1. Discuss about all the task from task 1 2. Tool for designing floor plan 3. Floor plan arrangement 4. Task distribution for each person		
Meeting MC	Poh Lok Yee		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1500	-	
Woo Cheng Shuan	1500	-	
Chau Ying Jia	1500	-	
Sabrina Heng Wei Qi	1520	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Suggest group name	All members suggested 4fabs as group name	All members
2	Shirt discussion on Task 1	All members read and understand the task given	All members
3	Software to use	Ying Jia suggested to use visual paradigm to draw floor plan	Ying Jia

4	Building design	<i>Ying Jia and Cheng Shuan suggested building design in rectangle shape</i> <i>Lok Yee suggested building design in octagon shape</i>	Ying Jia and Cheng Shuan
5	Decision on building design	<i>Sabrina, Ying Jia and Cheng Shuan drew the draft for rectangle building design</i> <i>Lok Yee drew the draft for octagon building design</i> <i>All members discuss on the advantages and disadvantages between the designs and make final decision on building design</i>	All members
6	Floor plan design	<i>All members discuss floor plan design</i> <i>Lok Yee suggested adding lift and accessible toilet for disability friendly</i> <i>Cheng Shuan suggested all 4 labs at the lower ground</i> <i>Sabrina suggested the size for Video conferencing room and hybrid classroom</i> <i>Ying Jia drew the draft of floor plan design</i>	All members

7	Task distribution	<i>Lok Yee distributes the tasks for each member by using spin tools</i> <i>Outcomes:</i> <i>Lok Yee is responsible for designing the student lounge and first floor; Ying Jia is responsible for designing the second floor and Embedded lab, Sabrina is responsible for designing Cisco Network lab and hybrid classroom while Cheng Shuan is responsible for designing the general lab and Video Conferencing room</i>	Lok Yee
8	Next Meeting	<i>8 November – All given tasks should be 90% completed</i>	All members
9	Meeting ended	1715	All members

MEETING MINUTES #2

DATE/TIME	8 Nov 2024 2pm	
LOCATION	KTDI MA7	
AGENDA	1. Finalize the floor plan 2. Learn packet tracer	
Meeting MC	Chau Ying Jia	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
Poh Lok Yee	1400	-
Woo Cheng Shuan	1400	-

Chau Ying Jia	1400	-	
Sabrina Heng Wei Qi	1400	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Sharing session on current progress	All members shared own progresses	All members
2	Comment and give suggestion for enhancement	Lok Yee reminded all labs must have 30 workstations Cheng Shuan suggested all the same elements using same scale Sabrina suggested the size of stairs and lift Ying Jia suggested both floor toilets using same design	All members
3	Design and edit each room and floor	All members edit the rooms following the suggestion and reminder from other group members	All members
4	Combination of each lab and room on both floors and finalize the floor plan design	All members put each of their designs inside the ground floor and first floor and finalize the design	All members
5	Next Meeting	14 November – All members should download and learn how to use packet tracer	All members

6	Meeting Ended	1800	All members
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MEETING MINUTES #3

DATE/TIME		14 Nov 2024 3pm	
LOCATION		KTDI MA7	
AGENDA		1. Use Packet Tracer to enhance the floor plan 2. Finalize the floor plan	
Meeting MC		Sabrina Heng Wei Qi	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Poh Lok Yee		1500	-
Woo Cheng Shuan		1500	-
Chau Ying Jia		1500	-
Sabrina Heng Wei Qi		1500	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discuss function packet tracer	All members shared their own understanding on packet tracer	All members
2	Use Packet Tracer to enhance the floor plan design	Each member use Packet Tracer to design their assigned room.	All members
3	Finalize the floor plan	Finalize the floor plan	All members
4	Meeting Ended	1845	All members

MEETING MINUTES #4

DATE/TIME	22 Nov 2024 10am		
LOCATION	Online Webex Meeting		
AGENDA	1. Discuss about all the task from task 2 2. Suggest possible questions for task 2 3. Select the most related questions from all suggestion 4. Task distribution for each person		
Meeting MC	Woo Cheng Shuan		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1000	-	
Woo Cheng Shuan	1000	-	
Chau Ying Jia	1000	-	
Sabrina Heng Wei Qi	1000	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discussion for task 2	All members read and understand requirements from task2.	All members
2	Suggest possible questions	Each member contributes 5-6 questions regarding the case study.	All members

3	Select the most related questions	<i>Sabrina lists out the questions from each member in the Word document to easier members to read in overall.</i> <i>All members read and understand the questions provided by other members.</i> <i>All members discussed the questions one by one and selected 20 most related questions that can be used in task 2.</i>	All members
4	Task distribution	<i>Lok Yee distributed the task to all members equally.</i> <i>Each member gets 5 questions.</i> <i>All members had to do research to find a solution for those questions with listing out the references.</i>	Lok Yee
5	Next Meeting	<i>26 November – All given tasks should be 95% completed</i>	All members
6	Meeting ended	<i>1300</i>	All members

MEETING MINUTES #5

DATE/TIME	26 Nov 2024 10am		
LOCATION	Online Webex Meeting		
AGENDA	1. Review the answers for each question 2. Finalize the most suitable questions to be used 3. Discuss on the feasibility assessment		
Meeting MC	Poh Lok Yee		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1000	-	
Woo Cheng Shuan	1000	-	
Chau Ying Jia	1000	-	
Sabrina Heng Wei Qi	1000	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Explain answer for each question	All members explained their responsive questions and answers one by one.	All members
2	Review the answer	All members review the questions and answer carefully	All members
3	Finalize the most related questions and answers	Sabrina suggested dropping the questions with answers that are not from any reference.	All members

		<i>Ying Jia suggested dropping the questions that are quite related.</i> <i>Cheng Shuan said that some answers are confusing.</i> <i>Lok Yee highlighted the question that can be used.</i> <i>Lok Yee finalizes 15 questions and answers.</i> <i>All members agree with the finalized list.</i>	
4	Discuss on the feasibility assessment	<i>Lok Yee explained the feasibility assessment to all members.</i> <i>Ying Jia suggested covering several aspects in this assessment.</i>	Lok Yee and Ying Jia
5	Meeting ended	<i>At 1230, the meeting ended after all discussions were done.</i>	All members

MEETING MINUTES #6

DATE/TIME	28 Dec 2024 4pm		
LOCATION	MA7		
AGENDA	1. Discuss about all the task from task 3 2. Suggest devices needed in task 3 3. Research about the price of all devices 4. Select the most suitable devices from all the choices 5. Task distribution for each person		
Meeting MC	Sabrina Heng Wei Qi		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	1600	-	
Woo Cheng Shuan	1600	-	
Chau Ying Jia	1600	-	
Sabrina Heng Wei Qi	1600	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discussion for task 3	All members read and understand requirements from task3.	All members

2	Suggest devices needed	<i>Sabrina and Ying Jia search and list out the suggested devices needed in ground floor of the building.</i> <i>Lok Yee and Cheng Shuan search and list out the suggested devices needed in first floor of the building.</i> <i>Ying Jia creates a table for listing out all the devices to easier members to read in overall.</i>	All members
3	Research about the prices of all devices	<i>Sabrina and Ying Jia search for the price of devices needed in ground floor of the building.</i> <i>Lok Yee and Cheng Shuan search for the price of devices needed in first floor of the building.</i>	All members
4	Select most suitable device	<i>All members discuss about the Pros and cons of each device and compare the price of each device.</i> <i>All members vote for the most suitable device to use in the building.</i>	All members

5	Task distribution	<i>Sabrina distributed the task to all members equally.</i> <i>All members had to do research on each device, list out the device quantity used, price and others</i>	Sabrina Heng Wei Qi
6	Next Meeting	<i>6 January – All given tasks should be 95% completed</i>	All members
7	Meeting ended	<i>1900</i>	All members

MEETING MINUTES #7

DATE/TIME	12 Jan 2025 7pm	
LOCATION	MA7	
AGENDA	1. Discuss about all the task from task 4 2. Discuss about the network distribution in our floor plan. 3. Discuss about the details in designing the network routes 4. Task distribution for members	
Meeting MC	Poh Lok Yee	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE

Poh Lok Yee	1900	-	
Woo Cheng Shuan	1900	-	
Chau Ying Jia	1900	-	
Sabrina Heng Wei Qi	1900	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discussion for task 4	<i>Poh Lok Yee brief us the information and rubrics in task 4 All members read and understand requirements from task3.</i>	Poh Lok Yee
2	Suggestion for network distribution	<i>Lok Yee and Ying Jia provide ideas in designing the network distribution in the floor plan. Sabrina and Cheng Shuan suggest the necessary details that need to be standardized.</i>	All members
3	Task Distribution	<i>Lok Yee distributed tasks to all members. Cheng Shuan and Sabrina was assigned to have an idea and plan the overall network of building Ying Jia was assigned to plan the details in the distribution of network for the first floor</i>	All members

		<i>Lok Yee will be planning for the details in ground floor network distribution</i> <i>All members will be calculating the usage of networking cables and stuff together</i>	
4	Meeting Ended	<i>Meeting ended after all discussions have done</i>	All members

MEETING MINUTES #8

DATE/TIME	19 Jan 2025 8pm	
LOCATION	Google meet (online)	
AGENDA	1. Review the current progress for each group member. 2. Finalize network distribution diagram 3. Check the calculation of the cable length needed	
Meeting MC	Woo Cheng Shuan	
ATTENDANCE		
NAME	TIME	REASON FOR ABSENCE
Poh Lok Yee	2000	-
Woo Cheng Shuan	2000	-
Chau Ying Jia	2000	-

Sabrina Heng Wei Qi		2000	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Review the current progress for each group member	Each group member present their parts	All members
2	Finalize network distribution diagram	All members give suggestion and discuss on what should be implemented and change before finalising the network distribution	All members
3	Calculation of the cable length needed	We calculate the total length of wire together and give suggestion and discuss about the calculation together	All members
4	Meeting Ended	Meeting ended after all discussions have done	All members

MEETING MINUTES #9

DATE/TIME	22 Jan 2025 8pm
LOCATION	Google Meeting
AGENDA	1. Discuss about all the task from task 5

	2. Discuss about the network division		
	3. <i>Complete task 5 together</i>		
Meeting MC	Chau Ying Jia		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
Poh Lok Yee	2000	-	
Woo Cheng Shuan	2000	-	
Chau Ying Jia	2000	-	
Sabrina Heng Wei Qi	2000	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE
1	Discussion for task 5	<i>Chau Ying Jia brief us the information and rubrics in task 5</i> <i>All members read and understand requirements from task 5.</i>	Chau Ying Jia
2	Suggestion for network division	<i>All members suggest their prefer subnetwork division and decide the one best suit our design</i>	All members
3	IP Addressing Scheme Overview	<i>All members examine how IP addresses are currently assigned to each work area.</i>	All members
4	Task Distribution	<i>Ying Jia distributed the tasks for all members.</i>	All members

5	Proceed Tasks 5 together	<i>All members work on Words together to complete task 5.</i>	All members
6	Meeting Ended	<i>Meeting ended after all discussions have done at 2300</i>	All members

9.3 Picture of Team Working on Project

Task 1



Figure 9.3.1 Discussion for Task 1 at MA7

Task 2

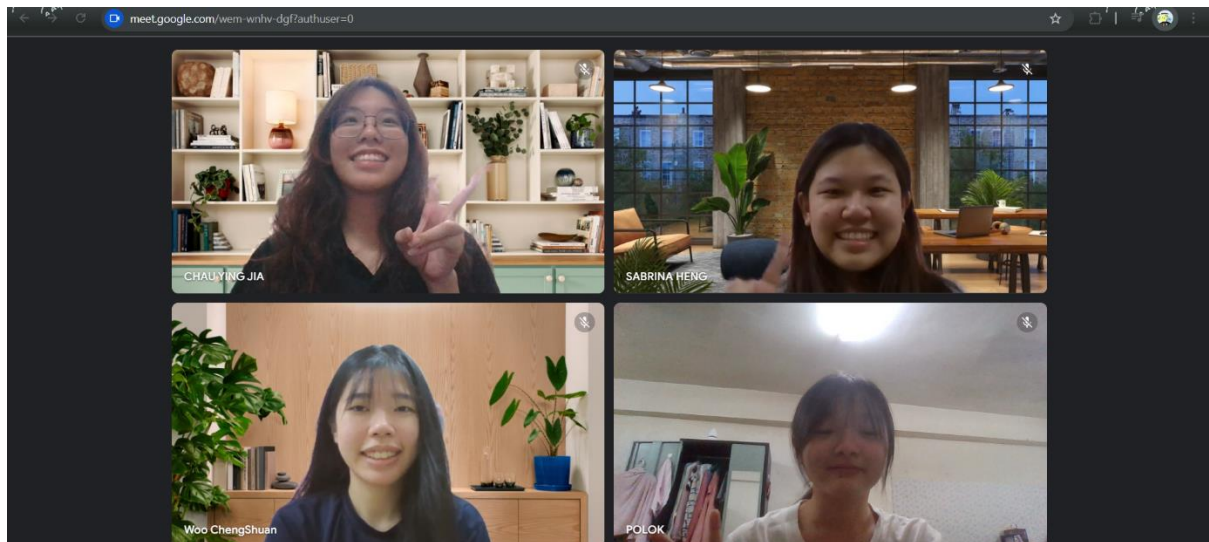


Figure 9.3.2 Online Discussion for Task 2

Task 3



Figure 9.3.3 Discussion for Task 3 at MA7

Task 4

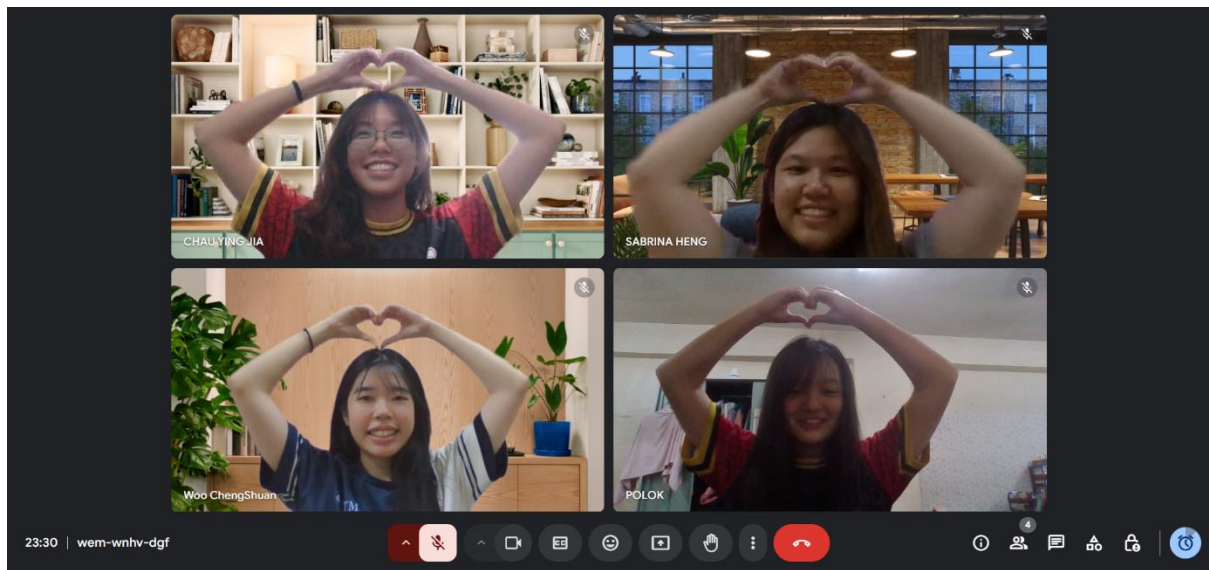


Figure 9.3.4 Online discussion at Google Meet for Task 4

Task 5



Figure 9.3.5 Online discussion for Task 5 at Google Meet

Task 6



Figure 9.3.6 Discussion for Task 6 at MA7